1) a)

$\sin^2 x + \cos^2 x \equiv 1 \Rightarrow -\sinh^2 x + \cosh^2 x \equiv 1$

$\therefore \cosh^2 x - \sinh^2 x \equiv 1$

1) b)

$1 + \cot^2 x \equiv \operatorname{cosec}^2 x \Rightarrow 1 + \frac{\cos^2 x}{\sin^2 x} \equiv \frac{1}{\sin^2 x} \Rightarrow 1 + \frac{\cosh^2 x}{-\sinh^2 x} \equiv \frac{1}{-\sinh^2 x}$

$\Rightarrow 1 - \coth^2 x \equiv -\operatorname{cosech}^2 x$

$\therefore \coth^2 x - 1 \equiv \operatorname{cosech}^2 x$

1) c)

$\tan^2 x + 1 \equiv \sec^2 x \Rightarrow \frac{\sin^2 x}{\cos^2 x} + 1 \equiv \frac{1}{\cos^2 x} \Rightarrow \frac{-\sinh^2 x}{\cosh^2 x} + 1 \equiv \frac{1}{\cosh^2 x}$

$\Rightarrow -\tanh^2 x + 1 \equiv \operatorname{sech}^2 x$

$\therefore 1 - \tanh^2 x \equiv \operatorname{sech}^2 x$

1) d)

$\sin(x + y) \equiv \sin x \cos y + \cos x \sin y \Rightarrow \sinh(x + y) \equiv \sinh x \cosh y + \cosh x \sinh y$

$\therefore \sinh(x + y) \equiv \sinh x \cosh y + \cosh x \sinh y$

1) e)

$\sin(x - y) \equiv \sin x \cos y - \cos x \sin y \Rightarrow \sinh(x - y) \equiv \sinh x \cosh y - \cosh x \sinh y$

$\therefore \sinh(x - y) \equiv \sinh x \cosh y - \cosh x \sinh y$

1) f)

$\cos(x + y) \equiv \cos x \cos y - \sin x \sin y \Rightarrow \cosh(x + y) \equiv \cosh x \cosh y - (-\sinh x \sinh y)$

$\therefore \cosh(x + y) \equiv \cosh x \cosh y + \sinh x \sinh y$

1) g)

$\cos(x - y) \equiv \cos x \cos y + \sin x \sin y \Rightarrow \cosh(x - y) \equiv \cosh x \cosh y + (-\sinh x \sinh y)$

$\therefore \cosh(x - y) \equiv \cosh x \cosh y - \sinh x \sinh y$

1) h)

$$\tan(x+y) \equiv \frac{\tan x+\tan y}{1-\tan x\tan y} \Rightarrow \frac{\sin(x+y)}{\cos(x+y)} \equiv \frac{\frac{\sin x}{\cos y}+\frac{\sin y}{\cos y}}{1-\frac{\sin x}{\cos x}\times\frac{\sin y}{\cos y}} \Rightarrow \frac{\sinh(x+y)}{\cosh(x+y)} \equiv \frac{\frac{\sinh x}{\cosh x}+\frac{\sinh y}{\cosh y}}{1-\frac{(-\sinh x\times\sinh y)}{\cosh x\times\cosh y}}$$

$$\therefore \tanh(x+y) \equiv \frac{\tanh x+\tanh y}{1+\tanh x\tanh y}$$

1) i)

$$\tan(x-y) \equiv \frac{\tan x-\tan y}{1+\tan x\tan y} \Rightarrow \frac{\sin(x-y)}{\cos(x-y)} \equiv \frac{\frac{\sin x}{\cos y}-\frac{\sin y}{\cos y}}{1+\frac{\sin x}{\cos x}\times\frac{\sin y}{\cos y}} \Rightarrow \frac{\sinh(x-y)}{\cosh(x-y)} \equiv \frac{\frac{\sinh x}{\cosh x}-\frac{\sinh y}{\cosh y}}{1+\frac{(-\sinh x\times\sinh y)}{\cosh x\times\cosh y}}$$

$$\therefore \tanh(x-y) \equiv \frac{\tanh x-\tanh y}{1-\tanh x\tanh y}$$

1) j)

$$\sin 2x \equiv 2\sin x\cos x \Rightarrow \sinh 2x \equiv 2\sinh x\cosh x$$

$$\therefore \sinh 2x \equiv 2\sinh x\cosh x$$

1) k)

$$\cos 2x \equiv \cos^2 x - \sin^2 x \Rightarrow \cosh 2x \equiv \cosh^2 x - (-\sinh^2 x)$$

$$\therefore \cosh 2x \equiv \cosh^2 x + \sinh^2 x$$

1) l)

$$\cos 2x \equiv 2\cos^2 x - 1 \Rightarrow \cosh 2x \equiv 2\cosh^2 x - 1$$

$$\therefore \cosh 2x \equiv 2\cosh^2 x - 1$$

1) m)

$$\cos 2x \equiv 1 - 2\sin^2 x \Rightarrow \cosh 2x \equiv 1 - 2(-\sinh^2 x)$$

$$\therefore \cosh 2x \equiv 1 + 2\sinh^2 x$$

1) n)

$$\tan 2x \equiv \frac{2\tan x}{1-\tan^2 x} \Rightarrow \frac{\sin 2x}{\cos 2x} \equiv \frac{2\times\frac{\sin x}{\cos x}}{1-\frac{\sin^2 x}{\cos^2 x}} \Rightarrow \frac{\sinh 2x}{\cosh 2x} \equiv \frac{2\times\frac{\sinh x}{\cosh x}}{1-\frac{(-\sinh^2 x)}{\cosh^2 x}}$$

$$\therefore \tanh 2x \equiv \frac{2\tanh x}{1+\tanh^2 x}$$

1) o)

$\sin 3x \equiv 3\sin x - 4\sin^3 x \Rightarrow \sin 3x \equiv 3\sin x - 4\sin x \times \sin^2 x$

$\Rightarrow \sinh 3x \equiv 3\sinh x - 4\sinh x \times (-\sinh^2 x)$

$\therefore \sinh 3x \equiv 3\sinh x + 4\sinh^3 x$

1) p)

$\cos 3x \equiv 4\cos^3 x - 3\cos x \Rightarrow \cosh 3x \equiv 4\cosh^3 x - 3\cosh x$

$\therefore \cosh 3x \equiv 4\cosh^3 x - 3\cosh x$

1) q)

$$\tan 3x \equiv \frac{3\tan x - \tan^3 x}{1 - 3\tan^2 x} \Rightarrow \frac{\sin 3x}{\cos 3x} \equiv \frac{3\times\frac{\sin x}{\cos x} - \frac{\sin^3 x}{\cos^3 x}}{1 - 3\times\frac{\sin^2 x}{\cos^2 x}} \Rightarrow \frac{\sinh 3x}{\cosh 3x} \equiv \frac{3\times\frac{\sinh x}{\cosh x} - \frac{\sinh x\times(-\sinh^2 x)}{\cosh^3 x}}{1 - 3\times\frac{(-\sinh^2 x)}{\cosh^2 x}}$$

$$\therefore \tanh 3x \equiv \frac{3\tanh x + \tanh^3 x}{1 + 3\tanh^2 x}$$

1) r)

$\sin 4x \equiv \cos x\,(4\sin x - 8\sin^3 x) \Rightarrow \sinh 4x \equiv \cosh x\left(4\sinh x - 8\sinh x \times (-\sinh^2 x)\right)$

$\therefore \sinh 4x \equiv \cosh x\,(4\sinh x + 8\sinh^3 x)$

1) s)

$\cos 4x \equiv 8\cos^4 x - 8\cos^2 x + 1 \Rightarrow \cosh 4x \equiv 8\cosh^4 x - 8\cosh^2 x + 1$

$\therefore \cosh 4x \equiv 8\cosh^4 x - 8\cosh^2 x + 1$

1) t)

$$\tan 4x \equiv \frac{4\tan x - 4\tan^3 x}{1 - 6\tan^2 x + \tan^4 x} \Rightarrow \frac{\sin 4x}{\cos 4x} \equiv \frac{4\times\frac{\sin x}{\cos x} - 4\times\frac{\sin^3 x}{\cos^3 x}}{1 - 6\times\frac{\sin^2 x}{\cos^2 x} + \frac{\sin^4 x}{\cos^4 x}}$$

$$\Rightarrow \frac{\sinh 4x}{\cosh 4x} \equiv \frac{4\times\frac{\sinh x}{\cosh x} - 4\times\frac{\sinh x\times(-\sinh^2 x)}{\cosh^3 x}}{1 - 6\times\frac{(-\sinh^2 x)}{\cosh^2 x} + \frac{(-\sinh^2 x)^2}{\cosh^4 x}}$$

$$\therefore \tanh 4x \equiv \frac{4\tanh x + 4\tanh^3 x}{1 + 6\tanh^2 x + \tanh^4 x}$$

2) a)

$$\frac{\sinh x}{\cosh^2 x} \equiv \frac{\sinh x}{\cosh x} \times \frac{1}{\cosh x} \equiv \tanh x\,\operatorname{sech} x$$

2) b)

$$\frac{\cosh x}{\sinh^2 x} \equiv \frac{\cosh x}{\sinh x} \times \frac{1}{\sinh x} \equiv \coth x \operatorname{cosech} x$$

3) a)

$$\operatorname{cosech} x \sinh x \cosh x$$
$$\equiv \frac{1}{\sinh x} \times \sinh x \times \cosh x$$
$$\equiv 1 \times \cosh x$$
$$\equiv \cosh x$$

3) b)

$$\tanh x \operatorname{sech} x \cosh x$$
$$\equiv \tanh x \times \frac{1}{\cosh x} \times \cosh x$$
$$\equiv \tanh x \times 1$$
$$\equiv \tanh x$$

3) c)

$$\tanh x \operatorname{cosech} x \coth x \sinh x$$
$$\equiv \tanh x \times \frac{1}{\sinh x} \times \frac{1}{\tanh x} \times \sinh x$$
$$\equiv \sinh x \times \frac{1}{\sinh x} \times \tanh x \times \frac{1}{\tanh x}$$
$$\equiv 1$$

4) a)

$$\frac{\operatorname{sech} x}{\operatorname{cosech} x} \equiv \frac{\frac{1}{\cosh x}}{\frac{1}{\sinh x}} \equiv \frac{\sinh x}{\cosh x} \equiv \tanh x$$

4) b)

$$\frac{\coth x}{\operatorname{sech} x} \equiv \frac{\frac{\cosh x}{\sinh x}}{\frac{1}{\cosh x}} \equiv \frac{\cosh^2 x}{\sinh x} \equiv \cosh^2 x \operatorname{cosech} x$$

4) c)

$$\frac{\operatorname{cosech} x}{\tanh x} \equiv \frac{\frac{1}{\sinh x}}{\frac{\sinh x}{\cosh x}} \equiv \frac{\cosh x}{\sinh^2 x} \equiv \cosh x \operatorname{cosech}^2 x$$

4) d)

$$\frac{\operatorname{sech} x}{\coth x} \equiv \frac{\frac{1}{\cosh x}}{\frac{\cosh x}{\sinh x}} \equiv \frac{\sinh x}{\cosh^2 x} \equiv \sinh x \operatorname{sech}^2 x$$

4) e)

$$\frac{\operatorname{sech}^2 x}{\coth x} \equiv \frac{\frac{1}{\cosh^2 x}}{\frac{\cosh x}{\sinh x}} \equiv \frac{\sinh x}{\cosh^3 x} \equiv \sinh x \operatorname{sech}^3 x$$

4) f)

$$\frac{\operatorname{cosech}^2 x \coth^2 x}{\operatorname{sech} x \sinh x} \equiv \frac{\frac{1}{\sinh^2 x} \times \frac{\cosh^2 x}{\sinh^2 x}}{\frac{1}{\cosh x} \times \sinh x} \equiv \frac{\frac{\cosh^2 x}{\sinh^4 x}}{\frac{\sinh x}{\cosh x}} \equiv \frac{\cosh^3 x}{\sinh^5 x} \equiv \cosh^3 x \operatorname{cosech}^5 x$$

5)

$$-\ln\left(k+\sqrt{k^2-1}\right)$$
$$= \ln\left(\left(k+\sqrt{k^2-1}\right)^{-1}\right)$$
$$= \ln\left(\frac{1}{k+\sqrt{k^2-1}}\right)$$
$$= \ln\left(\frac{1}{k+\sqrt{k^2-1}} \times \frac{k-\sqrt{k^2-1}}{k-\sqrt{k^2-1}}\right)$$
$$= \ln\left(\frac{k-\sqrt{k^2-1}}{k^2-(k^2-1)}\right)$$
$$= \ln\left(\frac{k-\sqrt{k^2-1}}{1}\right)$$
$$= \ln\left(k-\sqrt{k^2-1}\right)$$

6)

$$-\ln\left(k+\sqrt{k^2+1}\right)$$
$$=\ln\left(\left(k+\sqrt{k^2+1}\right)^{-1}\right)$$
$$=\ln\left(\frac{1}{k+\sqrt{k^2+1}}\right)$$
$$=\ln\left(\frac{1}{k+\sqrt{k^2+1}}\times\frac{k-\sqrt{k^2+1}}{k-\sqrt{k^2+1}}\right)$$
$$=\ln\left(\frac{k-\sqrt{k^2+1}}{k^2-(k^2+1)}\right)$$
$$=\ln\left(\frac{k-\sqrt{k^2+1}}{-1}\right)$$
$$=\ln\left(-k+\sqrt{k^2+1}\right)$$
$$\not\equiv \ln\left(k-\sqrt{k^2+1}\right)$$

7)

$$\cosh x = k$$
$$\Rightarrow \frac{e^x+e^{-x}}{2}=k$$
$$\Rightarrow e^x+e^{-x}=2k$$
$$\Rightarrow e^{2x}+1=2ke^x$$
$$\Rightarrow e^{2x}-2ke^x+1=0$$
$$\Rightarrow (e^x-k)^2-k^2+1=0$$
$$\Rightarrow (e^x-k)^2=k^2-1$$
$$\Rightarrow e^x-k=\pm\sqrt{k^2-1}$$
$$\Rightarrow e^x=k\pm\sqrt{k^2-1}$$
$$\Rightarrow x=\ln\left(k\pm\sqrt{k^2-1}\right)$$

8)

$\cosh x = 4$

$\Rightarrow \frac{e^x+e^{-x}}{2} = 4$

$\Rightarrow e^x + e^{-x} = 8$

$\Rightarrow e^{2x} + 1 = 8e^x$

$\Rightarrow e^{2x} - 8e^x + 1 = 0$

$\Rightarrow (e^x - 4)^2 - 16 + 1 = 0$

$\Rightarrow (e^x - 4)^2 = 15$

$\Rightarrow e^x - 4 = \pm\sqrt{15}$

$\Rightarrow e^x = 4 \pm \sqrt{15}$

$\Rightarrow x = \ln(4 \pm \sqrt{15})$

Now consider

$\ln(4 - \sqrt{15}) = \ln\left((4 - \sqrt{15}) \times \frac{4+\sqrt{15}}{4+\sqrt{15}}\right)$

$= \ln\left(\frac{16-15}{4+\sqrt{15}}\right)$

$= \ln\left(\frac{1}{4+\sqrt{15}}\right)$

$= \ln\left((4 + \sqrt{15})^{-1}\right)$

$= -\ln(4 + \sqrt{15})$

So $\ln(4 - \sqrt{15}) = -\ln(4 + \sqrt{15})$

So the solutions of $\cosh x = 4$ are $x = \ln(4 \pm \sqrt{15})$

which can also be written as $x = \pm\ln(4 + \sqrt{15})$

9) a)

$8\cosh^2 x + 7\sinh^2 x \equiv 8\cosh^2 x + 7(\cosh^2 x - 1) \equiv 15\cosh^2 x - 7$

9) b)

$3\sinh^2 x - 13\cosh^2 x \equiv 3(\cosh^2 x - 1) - 13\cosh^2 x \equiv -10\cosh^2 x - 3$

10)

$3\cosh^2 x - \sinh^2 x = 5$

$\Rightarrow 3\cosh^2 x - (\cosh^2 x - 1) = 5$

$\Rightarrow 2\cosh^2 x + 1 = 5$

$\Rightarrow \cosh^2 x = 2$

$\Rightarrow \cosh x = \pm\sqrt{2}$

$\cosh x = -\sqrt{2}$ has no real solutions

If $\cosh x = \sqrt{2}$, then $x = \pm\ln\left(\sqrt{2} + \sqrt{\left(\sqrt{2}\right)^2 - 1}\right) = \pm\ln\left(\sqrt{2} + 1\right)$

So $x = \ln\left(1 + \sqrt{2}\right)$ or $x = \ln\left(-1 + \sqrt{2}\right)$

11) a)

$\cosh^2 x - \sinh x - 7 = 0$

$\Rightarrow (\sinh^2 x + 1) - \sinh x - 7 = 0$

$\Rightarrow \sinh^2 x - \sinh x - 6 = 0$

$\Rightarrow (\sinh x - 3)(\sinh x + 2) = 0$

$\Rightarrow \sinh x = 3$ or $\sinh x = -2$

If $\sinh x = 3$, then $x = \ln\left(3 + \sqrt{3^2 + 1}\right) = \ln\left(3 + \sqrt{10}\right)$

If $\sinh x = -2$, then $x = \ln\left(-2 + \sqrt{(-2)^2 + 1}\right) = \ln\left(-2 + \sqrt{5}\right)$

So $x = \ln\left(3 + \sqrt{10}\right)$ or $x = \ln\left(-2 + \sqrt{5}\right)$

11) b)

$\sinh^2 x - 2\cosh x - 7 = 0$

$\Rightarrow (\cosh^2 x - 1) - 2\cosh x - 7 = 0$

$\Rightarrow \cosh^2 x - 2\cosh x - 8 = 0$

$\Rightarrow (\cosh x - 4)(\cosh x + 2) = 0$

$\Rightarrow \cosh x = 4$ or $\cosh x = -2$

$\cosh x = -2$ has no real solutions

If $\cosh x = 4$, then $x = \pm\ln\left(4 + \sqrt{4^2 - 1}\right) = \pm\ln\left(4 + \sqrt{15}\right)$

So $x = \ln\left(4 + \sqrt{15}\right)$ or $x = \ln\left(4 - \sqrt{15}\right)$

11) c)

$\cosh^2 x - 14\sinh x + 47 = 0$

$\Rightarrow (\sinh^2 x + 1) - 14\sinh x + 47 = 0$

$\Rightarrow \sinh^2 x - 14\sinh x + 48 = 0$

$\Rightarrow (\sinh x - 6)(\sinh x - 8) = 0$

$\Rightarrow \sinh x = 6$ or $\sinh x = 8$

If $\sinh x = 6$, then $x = \ln\left(6 + \sqrt{6^2 + 1}\right) = \ln\left(6 + \sqrt{37}\right)$

If $\sinh x = 8$, then $x = \ln\left(8 + \sqrt{8^2 + 1}\right) = \ln\left(8 + \sqrt{65}\right)$

So $x = \ln\left(6 + \sqrt{37}\right)$ or $x = \ln\left(8 + \sqrt{65}\right)$

11) d)

$2\sinh^2 x - 13\cosh x + 8 = 0$

$\Rightarrow 2(\cosh^2 x - 1) - 13\cosh x + 8 = 0$

$\Rightarrow 2\cosh^2 x - 13\cosh x + 6 = 0$

$\Rightarrow (2\cosh x - 1)(\cosh x - 6) = 0$

$\Rightarrow \cosh x = \frac{1}{2}$ or $\cosh x = 6$

If $\cosh x = \frac{1}{2}$, then there are no real solutions as $\cosh x \geq 1$ for all $x \in \mathbb{R}$

If $\cosh x = 6$, then $x = \pm\ln\left(6 + \sqrt{6^2 - 1}\right) = \pm\ln\left(6 + \sqrt{35}\right)$

So $x = \ln\left(6 + \sqrt{35}\right)$ or $x = \ln\left(6 - \sqrt{35}\right)$

11) e)

$3\cosh^2 x + 22\sinh x - 19 = 0$

$\Rightarrow 3(\sinh^2 x + 1) + 22\sinh x - 19 = 0$

$\Rightarrow 3\sinh^2 x + 22\sinh x - 16 = 0$

$\Rightarrow (3\sinh x - 2)(\sinh x + 8) = 0$

$\Rightarrow \sinh x = \frac{2}{3}$ or $\sinh x = -8$

If $\sinh x = \frac{2}{3}$, then $x = \ln\left(\frac{2}{3} + \sqrt{\left(\frac{2}{3}\right)^2 + 1}\right) = \ln\left(\frac{2}{3} + \frac{\sqrt{13}}{3}\right)$

If $\sinh x = -8$, then $x = \ln\left(-8 + \sqrt{(-8)^2 + 1}\right) = \ln\left(-8 + \sqrt{65}\right)$

So $x = \ln\left(\frac{2}{3} + \frac{\sqrt{13}}{3}\right)$ or $x = \ln\left(-8 + \sqrt{65}\right)$

11) f)

$4\sinh^2 x - 4\cosh x - 11 = 0$

$\Rightarrow 4(\cosh^2 x - 1) - 4\cosh x - 11 = 0$

$\Rightarrow 4\cosh^2 x - 4\cosh x - 15 = 0$

$\Rightarrow (2\cosh x + 3)(2\cosh x - 5) = 0$

$\Rightarrow \cosh x = -\frac{3}{2}$ or $\cosh x = \frac{5}{2}$

If $\cosh x = -\frac{3}{2}$, then there are no real solutions as $\cosh x \geq 1$ for all $x \in \mathbb{R}$

If $\cosh x = \frac{5}{2}$, then $x = \pm \ln\left(\frac{5}{2} + \sqrt{\left(\frac{5}{2}\right)^2 - 1}\right) = \pm \ln\left(\frac{5}{2} + \frac{\sqrt{21}}{2}\right)$

So $x = \ln\left(\frac{5}{2} + \frac{\sqrt{21}}{2}\right)$ or $x = \ln\left(\frac{5}{2} - \frac{\sqrt{21}}{2}\right)$

12) a)

$25\cosh^4 x - 40\cosh^2 x - 9 = 0$

$\Rightarrow (5\cosh^2 x + 1)(5\cosh^2 x - 9) = 0$

$\Rightarrow \cosh^2 x = -\frac{1}{5}$ or $\cosh^2 x = \frac{9}{5}$

$\cosh^2 x = -\frac{1}{5}$ has no real solutions as $\cosh^2 x \geq 1$

$\cosh^2 x = \frac{9}{5} \Rightarrow \cosh x = \pm\frac{3\sqrt{5}}{5}$

$\cosh x = -\frac{3\sqrt{5}}{5}$ has no real solutions as $\cosh x \geq 1$

If $\cosh x = \frac{3\sqrt{5}}{5}$, then $x = \pm \ln\left(\frac{3\sqrt{5}}{5} + \sqrt{\left(\frac{3\sqrt{5}}{5}\right)^2 - 1}\right) = \pm \ln\left(\frac{3\sqrt{5}}{5} + \frac{2\sqrt{5}}{5}\right) = \pm \ln\sqrt{5}$

So $x = \pm \ln\sqrt{5}$

12) b)

$36\cosh^4 x - 36\sinh^2 x - 91 = 0$

$\Rightarrow 36\cosh^4 x - 36(\cosh^2 x - 1) - 91 = 0$

$\Rightarrow 36\cosh^4 x - 36\cosh^2 x - 55 = 0$

$\Rightarrow (6\cosh^2 x + 5)(6\cosh^2 x - 11) = 0$

$\Rightarrow \cosh^2 x = -\frac{5}{6}$ or $\cosh^2 x = \frac{11}{6}$

$\cosh^2 x = -\frac{5}{6}$ has no real solutions as $\cosh^2 x \geq 1$

$\cosh^2 x = \frac{11}{6} \Rightarrow \cosh x = \pm\frac{\sqrt{66}}{6}$

$\cosh x = -\frac{\sqrt{66}}{6}$ has no real solutions as $\cosh x \geq 1$

If $\cosh x = \frac{\sqrt{66}}{6}$, then $x = \pm\ln\left(\frac{\sqrt{66}}{6} + \sqrt{\left(\frac{\sqrt{66}}{6}\right)^2 - 1}\right) = \pm\ln\left(\frac{\sqrt{66}}{6} + \frac{\sqrt{30}}{6}\right) = \pm\ln\left(\frac{\sqrt{66}+\sqrt{30}}{6}\right)$

So $x = \pm\ln\left(\frac{\sqrt{66}+\sqrt{30}}{6}\right)$

12) c)

$192\sinh^4 x - 172\cosh^2 x + 67 = 0$

$\Rightarrow 192\sinh^4 x - 172(1 + \sinh^2 x) + 67 = 0$

$\Rightarrow 192\sinh^4 x - 172\sinh^2 x - 105 = 0$

$\Rightarrow (16\sinh^2 x - 21)(12\sinh^2 x + 5) = 0$

$\Rightarrow \sinh^2 x = \frac{21}{16}$ or $\sinh^2 x = -\frac{5}{12}$

$\sinh^2 x = -\frac{5}{12}$ has no real solutions as $\sinh^2 x \geq 0$

$\sinh^2 x = \frac{21}{16} \Rightarrow \sinh x = \pm\frac{\sqrt{21}}{4}$

If $\sinh x = \frac{\sqrt{21}}{4}$, then $x = \ln\left(\frac{\sqrt{21}}{4} + \sqrt{\left(\frac{\sqrt{21}}{4}\right)^2 + 1}\right) = \ln\left(\frac{\sqrt{21}}{4} + \frac{\sqrt{37}}{4}\right) = \ln\left(\frac{\sqrt{21}+\sqrt{37}}{4}\right)$

If $\sinh x = -\frac{\sqrt{21}}{4}$, then $x = \ln\left(-\frac{\sqrt{21}}{4} + \sqrt{\left(-\frac{\sqrt{21}}{4}\right)^2 + 1}\right) = \ln\left(-\frac{\sqrt{21}}{4} + \frac{\sqrt{37}}{4}\right) =$

$\ln\left(\frac{\sqrt{37}-\sqrt{21}}{4}\right)$

So $x = \ln\left(\frac{\sqrt{21}+\sqrt{37}}{4}\right)$ or $x = \ln\left(\frac{\sqrt{37}-\sqrt{21}}{4}\right)$

13) a)

$\cosh^2 x - \sinh^2 x \equiv 1$

$\Rightarrow \frac{\cosh^2 x}{\sinh^2 x} - \frac{\sinh^2 x}{\sinh^2 x} \equiv \frac{1}{\sinh^2 x}$

$\Rightarrow \coth^2 x - 1 \equiv \operatorname{cosech}^2 x$

13) b)

$\cosh^2 x - \sinh^2 x \equiv 1$

$\Rightarrow \frac{\cosh^2 x}{\cosh^2 x} - \frac{\sinh^2 x}{\cosh^2 x} \equiv \frac{1}{\cosh^2 x}$

$\Rightarrow 1 - \tanh^2 x \equiv \operatorname{sech}^2 x$

14) a)

$2\operatorname{sech}^2 x - 3\tanh x = 0$

$\Rightarrow 2(1 - \tanh^2 x) - 3\tanh x = 0$

$\Rightarrow -2\tanh^2 x - 3\tanh x + 2 = 0$

$\Rightarrow 2\tanh^2 x + 3\tanh x - 2 = 0$

$\Rightarrow (2\tanh x - 1)(\tanh x + 2) = 0$

$\Rightarrow \tanh x = \frac{1}{2}$ or $\tanh x = -2$

If $\tanh x = \frac{1}{2}$, then $x = \frac{1}{2}\ln\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right) = \frac{1}{2}\ln 3$

If $\tanh x = -2$, then there are no real solutions as $-1 < \tanh x < 1$ for all $x \in \mathbb{R}$

So $x = \frac{1}{2}\ln 3$

14) b)

$3\tanh^2 x - 5\operatorname{sech} x - 1 = 0$

$\Rightarrow 3(1 - \operatorname{sech}^2 x) - 5\operatorname{sech} x - 1 = 0$

$\Rightarrow -3\operatorname{sech}^2 x - 5\operatorname{sech} x + 2 = 0$

$\Rightarrow 3\operatorname{sech}^2 x + 5\operatorname{sech} x - 2 = 0$

$\Rightarrow (3\operatorname{sech} x - 1)(\operatorname{sech} x + 2) = 0$

$\Rightarrow \operatorname{sech} x = \frac{1}{3}$ or $\operatorname{sech} x = -2$

$\Rightarrow \cosh x = 3$ or $\cosh x = -\frac{1}{2}$

If $\cosh x = 3$, then $x = \pm\ln\left(3 + \sqrt{3^2 - 1}\right) = \pm\ln\left(3 + 2\sqrt{2}\right)$

If $\cosh x = -\frac{1}{2}$, then there are no real solutions as $\cosh x \geq 1$ for all $x \in \mathbb{R}$

So $x = \ln\left(3 + 2\sqrt{2}\right)$ or $x = \ln\left(3 - 2\sqrt{2}\right)$

14) c)

$\operatorname{cosech}^2 x - 2\coth x - 7 = 0$

$\Rightarrow (\coth^2 x - 1) - 2\coth x - 7 = 0$

$\Rightarrow \coth^2 x - 2\coth x - 8 = 0$

$\Rightarrow (\coth x - 4)(\coth x + 2) = 0$

$\Rightarrow \coth x = 4$ or $\coth x = -2$

$\Rightarrow \tanh x = \frac{1}{4}$ or $\tanh x = -\frac{1}{2}$

If $\tanh x = \frac{1}{4}$, then $x = \frac{1}{2}\ln\left(\frac{1+\frac{1}{4}}{1-\frac{1}{4}}\right) = \frac{1}{2}\ln\frac{5}{3}$

If $\tanh x = -\frac{1}{2}$, then $x = \frac{1}{2}\ln\left(\frac{1+\left(-\frac{1}{2}\right)}{1-\left(-\frac{1}{2}\right)}\right) = \frac{1}{2}\ln\frac{1}{3}$

So $x = \frac{1}{2}\ln\frac{5}{3}$ or $x = \frac{1}{2}\ln\frac{1}{3}$

14) d)

$4\coth^2 x - 25\operatorname{cosech} x + 2 = 0$

$\Rightarrow 4(\operatorname{cosech}^2 x + 1) - 25\operatorname{cosech} x + 2 = 0$

$\Rightarrow 4\operatorname{cosech}^2 x - 25\operatorname{cosech} x + 6 = 0$

$\Rightarrow (4\operatorname{cosech} x - 1)(\operatorname{cosech} x - 6) = 0$

$\Rightarrow \operatorname{cosech} x = \frac{1}{4}$ or $\operatorname{cosech} x = 6$

$\Rightarrow \sinh x = 4$ or $\sinh x = \frac{1}{6}$

If $\sinh x = 4$, then $x = \ln\left(4 + \sqrt{4^2 + 1}\right) = \ln\left(4 + \sqrt{17}\right)$

If $\sinh x = \frac{1}{6}$, then $x = \ln\left(\frac{1}{6} + \sqrt{\left(\frac{1}{6}\right)^2 + 1}\right) = \ln\left(\frac{1}{6} + \sqrt{\frac{37}{6}}\right)$

So $x = \ln\left(4 + \sqrt{17}\right)$ or $x = \ln\left(\frac{1}{6} + \sqrt{\frac{37}{6}}\right)$

15) a)

$8\operatorname{sech}^2 x - 6\tanh x + 1 = 0$

$\Rightarrow 8(1 - \tanh^2 x) - 6\tanh x + 1 = 0$

$\Rightarrow 8 - 8\tanh^2 x - 6\tanh x + 1 = 0$

$\Rightarrow -8\tanh^2 x - 6\tanh x + 9 = 0$

$\Rightarrow 8\tanh^2 x + 6\tanh x - 9 = 0$

$\Rightarrow (4\tanh x - 3)(2\tanh x + 3) = 0$

$\Rightarrow \tanh x = \frac{3}{4}$ or $\tanh x = -\frac{3}{2}$

If $\tanh x = \frac{3}{4}$, then $x = \frac{1}{2}\ln\left(\frac{1+\frac{3}{4}}{1-\frac{3}{4}}\right) = \frac{1}{2}\ln 7$

$\tanh x = -\frac{3}{2}$ has no real solutions as $-1 < \tanh x < 1$

So $x = \frac{1}{2}\ln 7$

15) b)

$10\tanh^2 x - 21\operatorname{sech} x + 3 = 0$

$\Rightarrow 10(1 - \operatorname{sech}^2 x) - 21\operatorname{sech} x + 3 = 0$

$\Rightarrow 10 - 10\operatorname{sech}^2 x - 21\operatorname{sech} x + 3 = 0$

$\Rightarrow -10\operatorname{sech}^2 x - 21\operatorname{sech} x + 13 = 0$

$\Rightarrow 10\operatorname{sech}^2 x + 21\operatorname{sech} x - 13 = 0$

$\Rightarrow (5\operatorname{sech} x + 13)(2\operatorname{sech} x - 1) = 0$

$\Rightarrow \operatorname{sech} x = -\frac{13}{5}$ or $\operatorname{sech} x = \frac{1}{2}$

$\Rightarrow \cosh x = -\frac{5}{13}$ or $\cosh x = 2$

$\cosh x = -\frac{5}{13}$ has no real solutions as $\cosh x \geq 1$

If $\cosh x = 2$, then $x = \pm\ln\left(2 + \sqrt{2^2 - 1}\right) = \pm\ln\left(2 + \sqrt{3}\right)$

So $x = \ln\left(2 + \sqrt{3}\right)$ or $x = \ln\left(2 - \sqrt{3}\right)$

15) c)

$24\,\text{cosech}^2 x + 58\coth x + 59 = 0$

$\Rightarrow 24(\coth^2 x - 1) + 58\coth x + 59 = 0$

$\Rightarrow 24\coth^2 x - 24 + 58\coth x + 59 = 0$

$\Rightarrow 24\coth^2 x + 58\coth x + 35 = 0$

$\Rightarrow (6\coth x + 7)(4\coth x + 5) = 0$

$\Rightarrow \coth x = -\frac{7}{6}$ or $\coth x = -\frac{5}{4}$

If $\coth x = -\frac{7}{6}$, then $\tanh x = -\frac{6}{7} \Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\left(-\frac{6}{7}\right)}{1-\frac{6}{7}}\right) = \frac{1}{2}\ln\frac{1}{13}$

If $\coth x = -\frac{5}{4}$, then $\tanh x = -\frac{4}{5} \Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\left(-\frac{4}{5}\right)}{1-\left(-\frac{4}{5}\right)}\right) = \frac{1}{2}\ln\frac{1}{9}$

So $x = \frac{1}{2}\ln\left(\frac{1}{13}\right)$ or $x = \frac{1}{2}\ln\frac{1}{9}$

15) d)

$25\coth^2 x - 73\,\text{cosech}\,x = 31$

$\Rightarrow 25(\text{cosech}^2 x + 1) - 73\,\text{cosech}\,x = 31$

$\Rightarrow 25\,\text{cosech}^2 x - 73\,\text{cosech}\,x - 6 = 0$

$\Rightarrow (25\,\text{cosech}\,x + 2)(\text{cosech}\,x - 3) = 0$

$\Rightarrow \text{cosech}\,x = -\frac{2}{25}$ or $\text{cosech}\,x = 3$

If $\text{cosech}\,x = -\frac{2}{25}$, then $\sinh x = -\frac{25}{2} \Rightarrow x = \ln\left(-\frac{25}{2} + \sqrt{\left(-\frac{25}{2}\right)^2 + 1}\right) =$

$\ln\left(-\frac{25}{2} + \frac{\sqrt{629}}{2}\right)$

If $\text{cosech}\,x = 3$, then $\sinh x = \frac{1}{3} \Rightarrow x = \ln\left(\frac{1}{3} + \sqrt{\left(\frac{1}{3}\right)^2 + 1}\right) = \ln\left(\frac{1}{3} + \frac{\sqrt{10}}{3}\right)$

So $x = \ln\left(-\frac{25}{2} + \frac{\sqrt{629}}{2}\right)$ or $x = \ln\left(\frac{1}{3} + \frac{\sqrt{10}}{3}\right)$

16)

$6\tanh^2 x + 17\operatorname{sech} x - 18 = 0$

$\Rightarrow 6(1 - \operatorname{sech}^2 x) + 17\operatorname{sech} x - 18 = 0$

$\Rightarrow -6\operatorname{sech}^2 x + 17\operatorname{sech} x - 12 = 0$

$\Rightarrow 6\operatorname{sech}^2 x - 17\operatorname{sech} x + 12 = 0$

$\Rightarrow (3\operatorname{sech} x - 4)(2\operatorname{sech} x - 3) = 0$

$\Rightarrow \operatorname{sech} x = \frac{4}{3}$ or $\operatorname{sech} x = \frac{3}{2}$

However, $0 < \operatorname{sech} x \leq 1$, so $\operatorname{sech} x = \frac{4}{3}$ and $\operatorname{sech} x = \frac{3}{2}$ have no real solutions.

17) a)

$\sinh x = \operatorname{sech} x$

$\Rightarrow \sinh x = \frac{1}{\cosh x}$

$\Rightarrow \sinh x \cosh x = 1$

$\Rightarrow \frac{1}{2}\sinh 2x = 1$

$\Rightarrow \sinh 2x = 2$

$\Rightarrow 2x = \ln\left(2 + \sqrt{2^2 + 1}\right)$

$\Rightarrow x = \frac{1}{2}\ln\left(2 + \sqrt{5}\right)$

17) b)

$\tanh x = \operatorname{sech} x$

$\Rightarrow \frac{\sinh x}{\cosh x} = \frac{1}{\cosh x}$

$\Rightarrow \sinh x = 1$

$\Rightarrow x = \ln\left(1 + \sqrt{1^2 + 1}\right)$

$\Rightarrow x = \ln\left(1 + \sqrt{2}\right)$

17) c)

$\sinh x = 2\coth x$

$\Rightarrow \sinh x = 2 \times \frac{\cosh x}{\sinh x}$

$\Rightarrow \sinh^2 x = 2\cosh x$

$\Rightarrow \sinh^2 x - 2\cosh x = 0$

$\Rightarrow (\cosh^2 x - 1) - 2\cosh x = 0$

$\Rightarrow \cosh^2 x - 2\cosh x - 1 = 0$

$\Rightarrow (\cosh x - 1)^2 - 1 - 1 = 0$

$\Rightarrow (\cosh x - 1)^2 = 2$

$\Rightarrow \cosh x - 1 = \pm\sqrt{2}$

$\Rightarrow \cosh x = 1 \pm \sqrt{2}$

If $\cosh x = 1 + \sqrt{2}$, then $x = \pm \ln\left(1 + \sqrt{2} + \sqrt{(1+\sqrt{2})^2 - 1}\right)$

$= \pm \ln\left(1 + \sqrt{2} + \sqrt{2 + 2\sqrt{2}}\right)$

If $\cosh x = 1 - \sqrt{2}$, then there are no real solutions as $\cosh x \geq 1$ for all $x \in \mathbb{R}$

So $x = \ln\left(1 + \sqrt{2} + \sqrt{2 + 2\sqrt{2}}\right)$ or $x = \ln\left(1 + \sqrt{2} - \sqrt{2 + 2\sqrt{2}}\right)$

17) d)

$3\cosh x = \coth x$

$\Rightarrow 3\cosh x = \frac{\cosh x}{\sinh x}$

$\Rightarrow 3\sinh x \cosh x = \cosh x$

$\Rightarrow 3\sinh x \cosh x - \cosh x = 0$

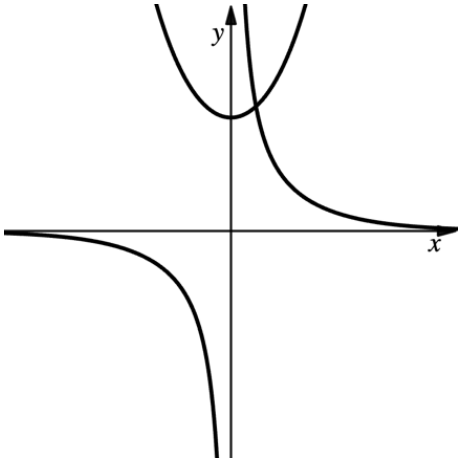
$\Rightarrow \cosh x\,(3\sinh x - 1) = 0$

$\Rightarrow \cosh x = 0$ or $\sinh x = \frac{1}{3}$

If $\cosh x = 0$, then there are no real solutions as $\cosh x \geq 1$ for all $x \in \mathbb{R}$

If $\sinh x = \frac{1}{3}$, then $x = \ln\left(\frac{1}{3} + \sqrt{\left(\frac{1}{3}\right)^2 + 1}\right) = \ln\left(\frac{1}{3} + \frac{\sqrt{10}}{3}\right)$

18) a)



18) b)

$2\cosh x = \operatorname{cosech} x$

$\Rightarrow 2\cosh x = \frac{1}{\sinh x}$

$\Rightarrow 2\sinh x \cosh x = 1$

$\Rightarrow \sinh 2x = 1$

$\Rightarrow 2x = \ln\left(1+\sqrt{1^2+1}\right)$

$\Rightarrow x = \frac{1}{2}\ln\left(1+\sqrt{2}\right)$

Let $y = 2\cosh\left(\frac{1}{2}\ln\left(1+\sqrt{2}\right)\right)$

$= 2\times\frac{e^{\frac{1}{2}\ln\left(1+\sqrt{2}\right)}+e^{-\frac{1}{2}\ln\left(1+\sqrt{2}\right)}}{2}$

$= e^{\frac{1}{2}\ln\left(1+\sqrt{2}\right)} + e^{-\frac{1}{2}\ln\left(1+\sqrt{2}\right)}$

$= e^{\ln\left(\left(1+\sqrt{2}\right)^{\frac{1}{2}}\right)} + e^{\ln\left(\left(1+\sqrt{2}\right)^{-\frac{1}{2}}\right)}$

$= \left(1+\sqrt{2}\right)^{\frac{1}{2}} + \left(1+\sqrt{2}\right)^{-\frac{1}{2}}$

$= \sqrt{1+\sqrt{2}} + \frac{1}{\sqrt{1+\sqrt{2}}}$

$= \frac{1+\sqrt{2}}{\sqrt{1+\sqrt{2}}} + \frac{1}{\sqrt{1+\sqrt{2}}}$

$= \frac{2+\sqrt{2}}{\sqrt{1+\sqrt{2}}}$

So the two graphs intersect at $\left(\frac{1}{2}\ln\left(1+\sqrt{2}\right), \frac{2+\sqrt{2}}{\sqrt{1+\sqrt{2}}}\right)$

Alternatively:

$2\cosh x = \operatorname{cosech} x$

$2\left(\frac{e^x+e^{-x}}{2}\right)=\frac{2}{e^x-e^{-x}}$

$(e^x+e^{-x})(e^x-e^{-x})=2$

$e^{2x}-e^{-2x}=2$

$e^{2x}-e^{-2x}-2=0$

$e^{4x}-2e^{2x}-1=0$

$(e^{2x}-1)^2-2=0$

$(e^{2x}-1)^2=2$

$e^{2x}-1=\pm\sqrt{2}$

$e^{2x}=1\pm\sqrt{2}$

$2x=\ln(1\pm\sqrt{2})$

$x=\frac{1}{2}\ln(1\pm\sqrt{2})$

Intersect at positive value:

so $x=\frac{1}{2}\ln(1+\sqrt{2})$

$y=2\cosh\left(\frac{1}{2}\ln(1+\sqrt{2})\right)=2\cosh\left(\ln\left(\sqrt{1+\sqrt{2}}\right)\right)$

$y=2\times\frac{e^{\ln\left(\sqrt{1+\sqrt{2}}\right)}+e^{-\ln\left(\sqrt{1+\sqrt{2}}\right)}}{2}$

$y=\sqrt{1+\sqrt{2}}+\frac{1}{\sqrt{1+\sqrt{2}}}$

$=\frac{1+\sqrt{2}+1}{\sqrt{1+\sqrt{2}}}=\frac{2+\sqrt{2}}{\sqrt{1+\sqrt{2}}}\times\frac{\sqrt{1+\sqrt{2}}}{\sqrt{1+\sqrt{2}}}$

$=\frac{(2+\sqrt{2})\sqrt{1+\sqrt{2}}}{1+\sqrt{2}}\times\frac{1-\sqrt{2}}{1-\sqrt{2}}=\frac{-\sqrt{2}\sqrt{1+\sqrt{2}}}{-1}$

$=\sqrt{2}\sqrt{1+\sqrt{2}}=\sqrt{2(1+\sqrt{2})}$

So the two graphs intersect at $\left(\frac{1}{2}\ln(1+\sqrt{2}),\sqrt{2(1+\sqrt{2})}\right)$

19) a)

$\sinh x = 3$

$\cosh^2 x - \sinh^2 x = 1$

$\Rightarrow \cosh^2 x - 3^2 = 1$

$\Rightarrow \cosh^2 x = 10$

$\Rightarrow \cosh x = \sqrt{10}$

19) b)

$\tanh x = \frac{\sinh x}{\cosh x}$

$\Rightarrow \tanh x = \frac{3}{\sqrt{10}} = \frac{3\sqrt{10}}{10}$

19) c)

$\coth x = \frac{\cosh x}{\sinh x} = \frac{\sqrt{10}}{3}$

19) d)

$\operatorname{cosech} x = \frac{1}{\sinh x} = \frac{1}{3}$

19) e)

$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{1}{\sqrt{10}} = \frac{\sqrt{10}}{10}$

20) a)

$1 - \tanh^2 x = \operatorname{sech}^2 x$

$\Rightarrow 1 - \tanh^2 x = \left(\frac{1}{\sqrt{2}}\right)^2$

$\Rightarrow \tanh^2 x = \frac{1}{2}$

$\Rightarrow \tanh x = \pm\frac{1}{\sqrt{2}}$

20) b)

$\cosh x = \frac{1}{\operatorname{sech} x} = \frac{1}{\frac{1}{\sqrt{2}}} = \sqrt{2}$

20) c)

$\cosh^2 x - \sinh^2 x = 1$

$\Rightarrow \left(\sqrt{2}\right)^2 - \sinh^2 x = 1$

$\Rightarrow \sinh^2 x = 1$

$\Rightarrow \sinh x = \pm 1$

20) d)

$\operatorname{cosech} x = \frac{1}{\sinh x} = \frac{1}{\pm 1} = \pm 1$

20) e)

$\coth x = \frac{1}{\tanh x} = \frac{1}{\pm\frac{1}{\sqrt{2}}} = \pm\sqrt{2}$

21) a)

$48\coth^2 x + 34\operatorname{cosech} x - 53 = 0$

$\Rightarrow 48(\operatorname{cosech}^2 x + 1) + 34\operatorname{cosech} x - 53 = 0$

$\Rightarrow 48\operatorname{cosech}^2 x + 34\operatorname{cosech} x - 5 = 0$

$\Rightarrow (6\operatorname{cosech} x + 5)(8\operatorname{cosech} x - 1) = 0$

$\Rightarrow \operatorname{cosech} x = -\frac{5}{6}$ or $\operatorname{cosech} x = \frac{1}{8}$

But $\operatorname{cosech} x = \frac{1}{8}$ only as $x > 0$

21) b)

$\sinh x = \frac{1}{\operatorname{cosech} x} = \frac{1}{\frac{1}{8}} = 8$

21) c)

$\cosh^2 x - \sinh^2 x = 1$

$\Rightarrow \cosh^2 x - 8^2 = 1$

$\Rightarrow \cosh^2 x = 65$

$\Rightarrow \cosh x = \pm\sqrt{65}$

But $\cosh x = \sqrt{65}$ as $\cosh x \geq 1$ for all real values of x

21) d)

$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{1}{\sqrt{65}} = \frac{\sqrt{65}}{65}$

21) e)

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{8}{\sqrt{65}} = \frac{8\sqrt{65}}{65}$$

21) f)

$$\coth x = \frac{1}{\tanh x} = \frac{1}{\frac{8\sqrt{65}}{65}} = \frac{65}{8\sqrt{65}} = \frac{\sqrt{65}}{8}$$

22) a)

$$\frac{\operatorname{sech}^2 x}{1-\tanh^2 x} \equiv \frac{\operatorname{sech}^2 x}{\operatorname{sech}^2 x} \equiv 1$$

22) b)

$$\frac{\coth^2 x-1}{\operatorname{cosech}^2 x} \equiv \frac{\operatorname{cosech}^2 x}{\operatorname{cosech}^2 x} \equiv 1$$

22) c)

$$\frac{1-\tanh^2 x}{\coth^2 x-1} \equiv \frac{\operatorname{sech}^2 x}{\operatorname{cosech}^2 x} \equiv \frac{\frac{1}{\cosh^2 x}}{\frac{1}{\sinh^2 x}} \equiv \frac{1}{\cosh^2 x} \times \frac{\sinh^2 x}{1} \equiv \frac{\sinh^2 x}{\cosh^2 x} \equiv \tanh^2 x$$

22) d)

$$\frac{3-3\coth^2 x}{8-8\tanh^2 x} \equiv \frac{-3(\coth^2 x-1)}{8(1-\tanh^2 x)} \equiv -\frac{3\operatorname{cosech}^2 x}{8\operatorname{sech}^2 x} \equiv -\frac{\frac{3}{\sinh^2 x}}{\frac{8}{\cosh^2 x}} \equiv -\frac{3}{\sinh^2 x} \times \frac{\cosh^2 x}{8} \equiv -\frac{3\cosh^2 x}{8\sinh^2 x}$$

$$\equiv -\frac{3}{8}\coth^2 x$$

23) a)

$\cosh^2 x - \sinh^2 x = 1$

$\Rightarrow \cosh^2(1+\ln 6) - \sinh^2(1+\ln 6) = 1$

$\Rightarrow \cosh^2(1+\ln 6) - \left(\frac{36e^2-1}{12e}\right)^2 = 1$

$\Rightarrow \cosh^2(1+\ln 6) = 1 + \frac{(36e^2-1)^2}{(12e)^2}$

$\Rightarrow \cosh^2(1+\ln 6) = \frac{(12e)^2+(36e^2-1)^2}{(12e)^2}$

$\Rightarrow \cosh^2(1+\ln 6) = \frac{144e^2+1296e^4-72e^2+1}{(12e)^2}$

$\Rightarrow \cosh^2(1+\ln 6) = \frac{1296e^4+72e^2+1}{(12e)^2}$

$\Rightarrow \cosh^2(1+\ln 6) = \frac{(36e^2+1)^2}{(12e)^2}$

$\Rightarrow \cosh(1+\ln 6) = \frac{36e^2+1}{12e}$

23) b)

$\tanh(1+\ln 6) = \frac{\sinh(1+\ln 6)}{\cosh(1+\ln 6)}$

$= \frac{\frac{36e^2-1}{12e}}{\frac{36e^2+1}{12e}}$

$= \frac{36e^2-1}{36e^2+1}$

24) a)

$p\left(\frac{5}{4}\right) = 12\left(\frac{5}{4}\right)^3 - 7\left(\frac{5}{4}\right)^2 - 42\left(\frac{5}{4}\right) + 40$

$= \frac{375}{16} - \frac{175}{16} - \frac{105}{2} + 40$

$= \frac{375}{16} - \frac{175}{16} - \frac{840}{16} + \frac{640}{16}$

$= \frac{200}{16} - \frac{840}{16} + \frac{640}{16}$

$= -\frac{640}{16} + \frac{640}{16} = 0$

$\therefore (4x-5)$ is a factor of $p(x)$

24) b)

$\mathrm{p}(x) = 12x^3 - 7x^2 - 42x + 40$

$= (4x - 5)(3x^2 + 2x - 8)$

$= (4x - 5)(3x - 4)(x + 2)$

24) c)

$12\operatorname{sech}^3\theta + 7\tanh^2\theta - 42\operatorname{sech}\theta + 33 = 0$

$\Rightarrow 12\operatorname{sech}^3\theta + 7(1 - \operatorname{sech}^2\theta) - 42\operatorname{sech}\theta + 33 = 0$

$\Rightarrow 12\operatorname{sech}^3\theta - 7\operatorname{sech}^2\theta - 42\operatorname{sech}\theta + 40 = 0$

$\Rightarrow (4\operatorname{sech}\theta - 5)(3\operatorname{sech}\theta - 4)(\operatorname{sech}\theta + 2) = 0$

$\Rightarrow \operatorname{sech}\theta = \frac{5}{4}$ or $\operatorname{sech}\theta = \frac{4}{3}$ or $\operatorname{sech}\theta = -2$

However, $0 < \operatorname{sech}\theta \leq 1$, so $\operatorname{sech}\theta = \frac{5}{4}$, $\operatorname{sech}\theta = \frac{4}{3}$ and $\operatorname{sech}\theta = -2$ have no real solutions.

25) a)

$$\sinh(\ln 3) = \frac{e^{\ln 3} - e^{-\ln 3}}{2} = \frac{e^{\ln 3} - e^{\ln\frac{1}{3}}}{2} = \frac{3 - \frac{1}{3}}{2} = \frac{4}{3}$$

$$\cosh(\ln 3) = \frac{e^{\ln 3} + e^{-\ln 3}}{2} = \frac{e^{\ln 3} + e^{\ln\frac{1}{3}}}{2} = \frac{3 + \frac{1}{3}}{2} = \frac{5}{3}$$

25) b)

$$\sinh(\ln 4) = \frac{e^{\ln 4} - e^{-\ln 4}}{2} = \frac{e^{\ln 4} - e^{\ln\frac{1}{4}}}{2} = \frac{4 - \frac{1}{4}}{2} = \frac{15}{8}$$

$$\cosh(\ln 4) = \frac{e^{\ln 4} + e^{-\ln 4}}{2} = \frac{e^{\ln 4} + e^{\ln\frac{1}{4}}}{2} = \frac{4 + \frac{1}{4}}{2} = \frac{17}{8}$$

25) c)

$\sinh(\ln 4 - \ln 3) = \sinh(\ln 4)\cosh(\ln 3) - \cosh(\ln 4)\sinh(\ln 3)$

$= \frac{15}{8} \times \frac{5}{3} - \frac{17}{8} \times \frac{4}{3} = \frac{25}{8} - \frac{17}{6} = \frac{7}{24}$

26) a)

$$\sinh(\ln 5) = \frac{e^{\ln 5} - e^{-\ln 5}}{2} = \frac{e^{\ln 5} - e^{\ln \frac{1}{5}}}{2} = \frac{5 - \frac{1}{5}}{2} = \frac{12}{5}$$

$$\cosh(\ln 5) = \frac{e^{\ln 5} + e^{-\ln 5}}{2} = \frac{e^{\ln 5} + e^{\ln \frac{1}{5}}}{2} = \frac{5 + \frac{1}{5}}{2} = \frac{13}{5}$$

26) b)

$$\sinh(\ln 6) = \frac{e^{\ln 6} - e^{-\ln 6}}{2} = \frac{e^{\ln 6} - e^{\ln \frac{1}{6}}}{2} = \frac{6 - \frac{1}{6}}{2} = \frac{35}{12}$$

$$\cosh(\ln 6) = \frac{e^{\ln 6} + e^{-\ln 6}}{2} = \frac{e^{\ln 6} + e^{\ln \frac{1}{6}}}{2} = \frac{6 + \frac{1}{6}}{2} = \frac{37}{12}$$

26) c)

$$\cosh(\ln 6 - \ln 5) = \cosh(\ln 6)\cosh(\ln 5) - \sinh(\ln 6)\sinh(\ln 5)$$
$$= \frac{37}{12} \times \frac{13}{5} - \frac{35}{12} \times \frac{12}{5} = \frac{481}{60} - 7 = \frac{61}{60}$$

27) a)

$$\tanh(\ln 2) = \frac{e^{\ln 2} - e^{-\ln 2}}{e^{\ln 2} + e^{-\ln 2}} = \frac{e^{\ln 2} - e^{\ln \frac{1}{2}}}{e^{\ln 2} + e^{\ln \frac{1}{2}}} = \frac{2 - \frac{1}{2}}{2 + \frac{1}{2}} = \frac{3}{5}$$

27) b)

$$\tanh(\ln 7) = \frac{e^{\ln 7} - e^{-\ln 7}}{e^{\ln 7} + e^{-\ln 7}} = \frac{e^{\ln 7} - e^{\ln \frac{1}{7}}}{e^{\ln 7} + e^{\ln \frac{1}{7}}} = \frac{7 - \frac{1}{7}}{7 + \frac{1}{7}} = \frac{24}{25}$$

27) c)

$$\tanh(\ln 2 + \ln 7) = \frac{\tanh(\ln 2) + \tanh(\ln 7)}{1 + \tanh(\ln 2)\tanh(\ln 7)} = \frac{\frac{3}{5} + \frac{24}{25}}{1 + \frac{3}{5} \times \frac{24}{25}} = \frac{195}{197}$$

28) a)

$$\sinh 3 \cosh 6 - \cosh 3 \sinh 6 = \sinh(3 - 6) = \sinh(-3)$$

28) b)

$$\sinh 4 \cosh 5 + \cos 4 \sinh 5 = \sinh(4 + 5) = \sinh 9$$

29) a)

$\cosh 3 \cosh 7 - \sinh 3 \sinh 7 = \cosh(3-7) = \cosh(-4)$

29) b)

$\cosh 8 \cosh 9 + \sinh 8 \sinh 9 = \cosh(8+9) = \cosh 17$

30) a)

$\cosh x = 2 \Rightarrow 2^2 - \sinh^2 x = 1 \Rightarrow \sinh^2 x = 3 \Rightarrow \sinh x = \sqrt{3}$

We must have that $\sinh x > 0$ as $\cosh x = 2 > 0$

$\cosh y = 3 \Rightarrow 3^2 - \sinh^2 y = 1 \Rightarrow \sinh^2 y = 8 \Rightarrow \sinh y = 2\sqrt{2}$

We must have that $\sinh y > 0$ as $\cosh y = 3 > 0$

$\therefore \cosh(x-y) = \cosh x \cosh y - \sinh x \sinh y$

$= 2 \times 3 - \sqrt{3} \times 2\sqrt{2} = 6 - 2\sqrt{6}$

30) b)

$\sinh(x-y) = \sinh x \cosh y - \sinh y \cosh x$

$= \sqrt{3} \times 3 - 2\sqrt{2} \times 2 = 3\sqrt{3} - 4\sqrt{2}$

31) a)

$\sinh(x + \ln 2) + \cosh(x - \ln 2)$

$= (\sinh x \cosh(\ln 2) + \cosh x \sinh(\ln 2)) + (\cosh x \cosh(\ln 2) - \sinh x \sinh(\ln 2))$

$= \frac{5}{4}\sinh x + \frac{3}{4}\cosh x + \frac{5}{4}\cosh x - \frac{3}{4}\sinh x$

$= \frac{1}{2}\sinh x + 2\cosh x$

31) b)

$\sinh(x - \ln 3) - \cosh(x + \ln 3)$

$= (\sinh x \cosh(\ln 3) - \cosh x \sinh(\ln 3)) - (\cosh x \cosh(\ln 3) + \sinh x \sinh(\ln 3))$

$= \frac{5}{3}\sinh x - \frac{4}{3}\cosh x - \frac{5}{3}\cosh x - \frac{4}{3}\sinh x$

$= \frac{1}{3}\sinh x - 3\cosh x$

31) c)

$\cosh(x - \ln 4) - \sinh(x - \ln 4)$

$= \cosh x \cosh(\ln 4) - \sinh x \sinh(\ln 4) - (\sinh x \cosh(\ln 4) - \cosh x \sinh(\ln 4))$

$= \frac{17}{8}\cosh x - \frac{15}{8}\sinh x - \frac{17}{8}\sinh x + \frac{15}{8}\cosh x$

$= 4\cosh x - 4\sinh x$

32) a)

$\sinh(x + \ln 2) = \cosh x$

$\Rightarrow \sinh x \cosh(\ln 2) + \cosh x \sinh(\ln 2) = \cosh x$

$\Rightarrow \frac{5}{4}\sinh x + \frac{3}{4}\cosh x = \cosh x$

$\Rightarrow \frac{5}{4}\sinh x = \frac{1}{4}\cosh x$

$\Rightarrow 5\sinh x = \cosh x$

$\Rightarrow \tanh x = \frac{1}{5}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\frac{1}{5}}{1-\frac{1}{5}}\right) = \frac{1}{2}\ln\frac{3}{2}$

32) b)

$\cosh(x - \ln 3) = \cosh x$

$\Rightarrow \cosh x \cosh(\ln 3) - \sinh x \sinh(\ln 3) = \cosh x$

$\Rightarrow \frac{5}{3}\cosh x - \frac{4}{3}\sinh x = \cosh x$

$\Rightarrow \frac{2}{3}\cosh x = \frac{4}{3}\sinh x$

$\Rightarrow \cosh x = 2\sinh x$

$\Rightarrow \tanh x = \frac{1}{2}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right) = \frac{1}{2}\ln 3$

32) c)

$\cosh(x - \ln 4) = \sinh x$

$\Rightarrow \cosh x \cosh(\ln 4) - \sinh x \sinh(\ln 4) = \sinh x$

$\Rightarrow \frac{17}{8}\cosh x - \frac{15}{8}\sinh x = \sinh x$

$\Rightarrow \frac{17}{8}\cosh x = \frac{23}{8}\sinh x$

$\Rightarrow 17\cosh x = 23\sinh x$

$\Rightarrow \tanh x = \frac{17}{23}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\frac{17}{23}}{1-\frac{17}{23}}\right) = \frac{1}{2}\ln\frac{20}{3}$

33) a)

$\sinh(x + \ln 2) = \cosh(x - \ln 4)$

$\Rightarrow \sinh x \cosh(\ln 2) + \cosh x \sinh(\ln 2) = \cosh x \cosh(\ln 4) - \sinh x \sinh(\ln 4)$

$\Rightarrow \frac{5}{4}\sinh x + \frac{3}{4}\cosh x = \frac{17}{8}\cosh x - \frac{15}{8}\sinh x$

$\Rightarrow \frac{25}{8}\sinh x = \frac{11}{8}\cosh x$

$\Rightarrow 25\sinh x = 11\cosh x$

$\Rightarrow \tanh x = \frac{11}{25}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\frac{11}{25}}{1-\frac{11}{25}}\right) = \frac{1}{2}\ln\frac{18}{7}$

33) b)

$\cosh(x - \ln 9) = \sinh(x + \ln 3)$

$\Rightarrow \cosh x \cosh(\ln 9) - \sinh x \sinh(\ln 9) = \sinh x \cosh(\ln 3) + \cosh x \sinh(\ln 3)$

$\Rightarrow \frac{41}{9}\cosh x - \frac{40}{9}\sinh x = \frac{5}{3}\sinh x + \frac{4}{3}\cosh x$

$\Rightarrow \frac{29}{9}\cosh x = \frac{55}{9}\sinh x$

$\Rightarrow 29\cosh x = 55\sinh x$

$\Rightarrow \tanh x = \frac{29}{55}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\frac{29}{55}}{1-\frac{29}{55}}\right) = \frac{1}{2}\ln\frac{42}{13}$

33) c)

$\cosh(x + 3\ln 2) = \cosh(x - \ln 20)$

$\Rightarrow \cosh x \cosh(3\ln 2) + \sinh x \sinh(3\ln 2) = \cosh x \cosh(\ln 20) - \sinh x \sinh(\ln 20)$

$\Rightarrow \frac{65}{16}\cosh x + \frac{63}{16}\sinh x = \frac{401}{40}\cosh x - \frac{399}{40}\sinh x$

$\Rightarrow \frac{1113}{80}\sinh x = \frac{477}{80}\cosh x$

$\Rightarrow 1113\sinh x = 477\cosh x$

$\Rightarrow \tanh x = \frac{477}{1113} = \frac{3}{7}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\frac{3}{7}}{1-\frac{3}{7}}\right) = \frac{1}{2}\ln\frac{5}{2}$

34) a)

LHS: $\coth x + \cosh x$

$\equiv \frac{\cosh x}{\sinh x} + \frac{\cosh x}{1}$

$\equiv \frac{\cosh x + \sinh x \cosh x}{\sinh x}$

$\equiv \frac{\cosh x(1+\sinh x)}{\sinh x}$

$\equiv \frac{\cosh x}{\sinh x} \times (1 + \sinh x)$

$\equiv \coth x\,(1 + \sinh x)$:RHS

34) b)

LHS: $\coth x + \operatorname{sech} x$

$\equiv \frac{\cosh x}{\sinh x} + \frac{1}{\cosh x}$

$\equiv \frac{\cosh^2 x + \sinh x}{\sinh x \cosh x}$

$\equiv \frac{1}{\sinh x} \times \frac{1}{\cosh x} \times (\cosh^2 x + \sinh x)$

$\equiv \operatorname{cosech} x \operatorname{sech} x\,(\sinh x + \cosh^2 x)$:RHS

34) c)

LHS: $\coth x - \tanh x$

$\equiv \frac{\cosh x}{\sinh x} - \frac{\sinh x}{\cosh x}$

$\equiv \frac{\cosh^2 x - \sinh^2 x}{\sinh x \cosh x}$

$\equiv \frac{1}{\sinh x \cosh x}$

$\equiv \frac{1}{\sinh x} \times \frac{1}{\cosh x}$

$\equiv \operatorname{cosech} x \operatorname{sech} x$:RHS

35) a)

LHS: $\cosh x - \operatorname{sech} x$

$\equiv \frac{\cosh x}{1} - \frac{1}{\cosh x}$

$\equiv \frac{\cosh^2 x - 1}{\cosh x}$

$\equiv \frac{\sinh^2 x}{\cosh x}$

$\equiv \sinh x \times \frac{\sinh x}{\cosh x}$

$\equiv \sinh x \tanh x$:RHS

35) b)

LHS: $\operatorname{cosech} x + \sinh x$

$\equiv \frac{1}{\sinh x} + \frac{\sinh x}{1}$

$\equiv \frac{1 + \sinh^2 x}{\sinh x}$

$\equiv \frac{\cosh^2 x}{\sinh x}$

$\equiv \cosh x \times \frac{\cosh x}{\sinh x}$

$\equiv \cosh x \coth x$: RHS

35) c)

RHS: $\tanh x\,(\coth^2 x - 1)$

$\equiv \coth x - \tanh x$

$\equiv \frac{\cosh x}{\sinh x} - \frac{\sinh x}{\cosh x}$

$\equiv \frac{\cosh^2 x - \sinh^2 x}{\sinh x \cosh x}$

$\equiv \frac{1}{\sinh x \cosh x}$

$\equiv \frac{1}{\sinh x} \times \frac{1}{\cosh x}$

$\equiv \operatorname{cosech} x \operatorname{sech} x$:LHS

35) d)

LHS: $\operatorname{cosech} x\,(\sinh x \tanh x - 1)$

$\equiv \frac{1}{\sinh x} \times \left(\sinh x \times \frac{\sinh x}{\cosh x} - 1\right)$

$\equiv \frac{1}{\sinh x} \times \left(\frac{\sinh^2 x}{\cosh x} - 1\right)$

$\equiv \frac{\sinh x}{\cosh x} - \frac{1}{\sinh x}$

$\equiv \tanh x - \operatorname{cosech} x$:RHS

36) a)

$8 \sinh 2x \cosh 2x \equiv 4(2 \sinh 2x \cosh 2x) \equiv 4 \sinh(2 \times 2x) \equiv 4 \sinh 4x$

36) b)

$2 \cosh^2 5x - 1 \equiv \cosh(2 \times 5x) \equiv \cosh 10x$

36) c)

$1 + 2 \sinh^2 3x \equiv \cosh(2 \times 3x) \equiv \cosh 6x$

36) d)

$9 \cosh^2 \frac{1}{4}x + 9 \sinh^2 \frac{1}{4}x \equiv 9\left(\cosh^2 \frac{1}{4}x + \sinh^2 \frac{1}{4}x\right) \equiv 9 \cosh\left(2 \times \frac{1}{4}x\right) \equiv 9 \cosh \frac{1}{2}x$

37) a)

$\sinh 2\theta - 2\sinh\theta = 0$

$\Rightarrow 2\sinh\theta\cosh\theta - 2\sinh\theta = 0$

$\Rightarrow \sinh\theta\,(\cosh\theta - 1) = 0$

$\Rightarrow \sinh\theta = 0$ or $\cosh\theta = 1$

If $\sinh\theta = 0$, then $\theta = \ln\left(0 + \sqrt{0^2 + 1}\right) = \ln 1 = 0$

If $\cosh\theta = 1$, then $\theta = \ln\left(1 + \sqrt{1^2 - 1}\right) = \ln 1 = 0$

So $\theta = 0$ is the only real solution.

37) b)

$\sinh 2\theta = 3\sinh\theta$

$\Rightarrow 2\sinh\theta\cosh\theta - 3\sinh\theta = 0$

$\Rightarrow \sinh\theta\,(2\cosh\theta - 3) = 0$

$\Rightarrow \sinh\theta = 0$ or $\cosh\theta = \frac{3}{2}$

If $\sinh\theta = 0$, then $\theta = \ln\left(0 + \sqrt{0^2 + 1}\right) = \ln 1 = 0$

If $\cosh\theta = \frac{3}{2}$, then $\theta = \pm\ln\left(\frac{3}{2} + \sqrt{\left(\frac{3}{2}\right)^2 - 1}\right) = \pm\ln\left(\frac{3}{2} + \frac{\sqrt{5}}{2}\right)$

So $\theta = 0$, $\theta = \ln\left(\frac{3}{2} + \frac{\sqrt{5}}{2}\right)$ or $\theta = \ln\left(\frac{3}{2} - \frac{\sqrt{5}}{2}\right)$

37) c)

$5\sinh 2\theta = 6\cosh\theta$

$\Rightarrow 10\sinh\theta\cosh\theta - 6\cosh\theta = 0$

$\Rightarrow \cosh\theta\,(5\sinh\theta - 3) = 0$

$\Rightarrow \cosh\theta = 0$ or $\sinh\theta = \frac{3}{5}$

$\cosh\theta = 0$ has no real solutions

If $\sinh\theta = \frac{3}{5}$, then $\theta = \ln\left(\frac{3}{5} + \sqrt{\left(\frac{3}{5}\right)^2 + 1}\right) = \ln\left(\frac{3}{5} + \frac{\sqrt{34}}{5}\right)$

So $\theta = \ln\left(\frac{3}{5} + \frac{\sqrt{34}}{5}\right)$

37) d)

$8\cosh\theta - 3\sinh 2\theta = 0$

$\Rightarrow 8\cosh\theta - 6\sinh\theta\cosh\theta = 0$

$\Rightarrow \cosh\theta\,(4 - 3\sinh\theta) = 0$

$\Rightarrow \cosh\theta = 0$ or $\sinh\theta = \frac{4}{3}$

$\cosh\theta = 0$ has no real solutions

If $\sinh\theta = \frac{4}{3}$, then $\theta = \ln\left(\frac{4}{3} + \sqrt{\left(\frac{4}{3}\right)^2 + 1}\right) = \ln\left(\frac{4}{3} + \frac{5}{3}\right) = \ln 3$

So $\theta = \ln 3$

38) a)

$2\cosh 2\theta = 7\cosh\theta$

$\Rightarrow 2\cosh 2\theta - 7\cosh\theta = 0$

$\Rightarrow 2(2\cosh^2\theta - 1) - 7\cosh\theta = 0$

$\Rightarrow 4\cosh^2\theta - 7\cosh\theta - 2 = 0$

$\Rightarrow \cosh\theta = -\frac{1}{4}$ or $\cosh\theta = 2$

$\cosh\theta = -\frac{1}{4}$ has no real solutions

If $\cosh\theta = 2$, then $\theta = \pm\ln\left(2 + \sqrt{2^2 - 1}\right) = \pm\ln\left(2 + \sqrt{3}\right)$

So $\theta = \ln\left(2 + \sqrt{3}\right)$ or $\theta = \ln\left(2 - \sqrt{3}\right)$

38) b)

$3\cosh 2\theta + 11\sinh\theta = 0$

$\Rightarrow 3(1 + 2\sinh^2\theta) + 11\sinh\theta = 0$

$\Rightarrow 6\sinh^2\theta + 11\sinh\theta + 3 = 0$

$\Rightarrow \sinh\theta = -\frac{3}{2}$ or $\sinh\theta = -\frac{1}{3}$

If $\sinh\theta = -\frac{3}{2}$, then $\theta = \ln\left(-\frac{3}{2} + \sqrt{\left(-\frac{3}{2}\right)^2 + 1}\right) = \ln\left(-\frac{3}{2} + \frac{\sqrt{13}}{2}\right)$

If $\sinh\theta = -\frac{1}{3}$, then $\theta = \ln\left(-\frac{1}{3} + \sqrt{\left(-\frac{1}{3}\right)^2 + 1}\right) = \ln\left(-\frac{1}{3} + \frac{\sqrt{10}}{3}\right)$

So $\theta = \ln\left(-\frac{3}{2} + \frac{\sqrt{13}}{2}\right)$ or $\theta = \ln\left(-\frac{1}{3} + \frac{\sqrt{10}}{3}\right)$

38) c)

$3\sinh\theta - \cosh 2\theta = 0$

$\Rightarrow 3\sinh\theta - (1 + 2\sinh^2\theta) = 0$

$\Rightarrow -2\sinh^2\theta + 3\sinh\theta - 1 = 0$

$\Rightarrow \sinh\theta = \frac{1}{2}$ or $\sinh\theta = 1$

If $\sinh\theta = \frac{1}{2}$, then $\theta = \ln\left(\frac{1}{2} + \sqrt{\left(\frac{1}{2}\right)^2 + 1}\right) = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$

If $\sinh\theta = 1$, then $\theta = \ln\left(1 + \sqrt{1^2 + 1}\right) = \ln\left(1 + \sqrt{2}\right)$

So $\theta = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$ or $\theta = \ln\left(1 + \sqrt{2}\right)$

38) d)

$\cosh 2\theta - 4\cosh\theta = 0$

$\Rightarrow (2\cosh^2\theta - 1) - 4\cosh\theta = 0$

$\Rightarrow 2\cosh^2\theta - 4\cosh\theta - 1 = 0$

$\Rightarrow \cosh\theta = \frac{2 \pm \sqrt{6}}{2}$

If $\cosh\theta = \frac{2+\sqrt{6}}{2}$, then $\theta = \pm\ln\left(\frac{2+\sqrt{6}}{2} + \sqrt{\left(\frac{2+\sqrt{6}}{2}\right)^2 - 1}\right) = \pm\ln\left(\frac{2+\sqrt{6}}{2} + \frac{\sqrt{6+4\sqrt{6}}}{2}\right) =$

$\pm\ln\left(\frac{2+\sqrt{6}}{2} + \frac{\sqrt{6+4\sqrt{6}}}{2}\right)$

So $\theta = \ln\left(\frac{2+\sqrt{6}}{2} + \frac{\sqrt{6+4\sqrt{6}}}{2}\right)$ or $\theta = \ln\left(\frac{2+\sqrt{6}}{2} - \frac{\sqrt{6+4\sqrt{6}}}{2}\right)$

39) a)

$\cosh 2\theta + 2\sinh\theta - 5 = 0$

$\Rightarrow (1 + 2\sinh^2\theta) + 2\sinh\theta - 5 = 0$

$\Rightarrow 2\sinh^2\theta + 2\sinh\theta - 4 = 0$

$\Rightarrow \sinh^2\theta + \sinh\theta - 2 = 0$

$\Rightarrow (\sinh\theta + 2)(\sinh\theta - 1) = 0$

$\Rightarrow \sinh\theta = -2$ or $\sinh\theta = 1$

If $\sinh\theta = -2$, then $\theta = \ln\left(-2 + \sqrt{(-2)^2 + 1}\right) = \ln\left(-2 + \sqrt{5}\right)$

If $\sinh\theta = 1$, then $\theta = \ln\left(1 + \sqrt{1^2 + 1}\right) = \ln\left(1 + \sqrt{2}\right)$

So $\theta = \ln\left(-2 + \sqrt{5}\right)$ or $\theta = \ln\left(1 + \sqrt{2}\right)$

39) b)

$5\cosh 2\theta - 3\sinh\theta = 5$

$\Rightarrow 5(1 + 2\sinh^2\theta) - 3\sinh\theta = 5$

$\Rightarrow 5 + 10\sinh^2\theta - 3\sinh\theta - 5 = 0$

$\Rightarrow 10\sinh^2\theta - 3\sinh\theta = 0$

$\Rightarrow \sinh\theta\,(10\sinh\theta - 3) = 0$

$\Rightarrow \sinh\theta = 0$ or $\sinh\theta = \frac{3}{10}$

If $\sinh\theta = 0$, then $\theta = \ln\left(0 + \sqrt{0^2 + 1}\right) = \ln 1 = 0$

If $\sinh\theta = \frac{3}{10}$, then $\theta = \ln\left(\frac{3}{10} + \sqrt{\left(\frac{3}{10}\right)^2 + 1}\right) = \ln\left(\frac{3}{10} + \frac{\sqrt{109}}{10}\right)$

So $\theta = 0$ or $\theta = \ln\left(\frac{3}{10} + \frac{\sqrt{109}}{10}\right)$

39) c)

$8\cosh\theta - 3\cosh 2\theta = 3$

$\Rightarrow 8\cosh\theta - 3(2\cosh^2\theta - 1) - 3 = 0$

$\Rightarrow 8\cosh\theta - 6\cosh^2\theta + 3 - 3 = 0$

$\Rightarrow -6\cosh^2\theta + 8\cosh\theta = 0$

$\Rightarrow 3\cosh^2\theta - 4\cosh\theta = 0$

$\Rightarrow \cosh\theta\,(3\cosh\theta - 4) = 0$

$\Rightarrow \cosh\theta = 0$ or $\cosh\theta = \frac{4}{3}$

$\cosh\theta = 0$ has no real solutions

If $\cosh\theta = \frac{4}{3}$, then $\theta = \pm\ln\left(\frac{4}{3} + \sqrt{\left(\frac{4}{3}\right)^2 - 1}\right) = \pm\ln\left(\frac{4}{3} + \frac{\sqrt{7}}{3}\right)$

So $\theta = \ln\left(\frac{4}{3} + \frac{\sqrt{7}}{3}\right)$ or $\theta = \ln\left(\frac{4}{3} - \frac{\sqrt{7}}{3}\right)$

39) d)

$5\cosh\theta - 12\cosh 2\theta + 17 = 0$

$\Rightarrow 5\cosh\theta - 12(2\cosh^2\theta - 1) + 17 = 0$

$\Rightarrow 5\cosh\theta - 24\cosh^2\theta + 12 + 17 = 0$

$\Rightarrow -24\cosh^2\theta + 5\cosh\theta + 29 = 0$

$\Rightarrow 24\cosh^2\theta - 5\cosh\theta - 29 = 0$

$\Rightarrow (24\cosh\theta - 29)(\cosh\theta + 1) = 0$

$\Rightarrow \cosh\theta = \frac{29}{24}$ or $\cosh\theta = -1$

If $\cosh\theta = \frac{29}{24}$, then $\theta = \pm\ln\left(\frac{29}{24} + \sqrt{\left(\frac{29}{24}\right)^2 - 1}\right) = \pm\ln\left(\frac{29}{24} + \frac{\sqrt{265}}{24}\right)$

$\cosh\theta = -1$ has no real solutions

So $\theta = \ln\left(\frac{29}{24} + \frac{\sqrt{265}}{24}\right)$ or $\theta = \ln\left(\frac{29}{24} - \frac{\sqrt{265}}{24}\right)$

40) a)

$2\sinh 2\theta\,\text{cosech}\,\theta = 5$

$\Rightarrow 2 \times 2\sinh\theta\cosh\theta \times \frac{1}{\sinh\theta} = 5$

$\Rightarrow 4\cosh\theta = 5$

$\Rightarrow \cosh\theta = \frac{5}{4}$

$\Rightarrow \theta = \pm\ln\left(\frac{5}{4} + \sqrt{\left(\frac{5}{4}\right)^2 - 1}\right) = \pm\ln\left(\frac{5}{4} + \frac{3}{4}\right) = \pm\ln 2$

So $\theta = \pm\ln 2$

40) b)

$5\sinh 2\theta\,\text{sech}\,\theta = 7$

$\Rightarrow 5 \times 2\sinh\theta\cosh\theta \times \frac{1}{\cosh\theta} = 7$

$\Rightarrow 10\sinh\theta = 7$

$\Rightarrow \sinh\theta = \frac{7}{10}$

$\Rightarrow \theta = \ln\left(\frac{7}{10} + \sqrt{\left(\frac{7}{10}\right)^2 + 1}\right) = \ln\left(\frac{7}{10} + \frac{\sqrt{149}}{10}\right)$

So $\theta = \ln\left(\frac{7}{10} + \frac{\sqrt{149}}{10}\right)$

40) c)

$\cosh 2\theta \operatorname{cosech} \theta = 3$

$\Rightarrow (1 + 2\sinh^2 \theta) \times \frac{1}{\sinh \theta} = 3$

$\Rightarrow 1 + 2\sinh^2 \theta = 3\sinh \theta$

$\Rightarrow 2\sinh^2 \theta - 3\sinh \theta + 1 = 0$

$\Rightarrow \sinh \theta = \frac{1}{2}$ or $\sinh \theta = 1$

If $\sinh \theta = \frac{1}{2}$, then $\theta = \ln\left(\frac{1}{2} + \sqrt{\left(\frac{1}{2}\right)^2 + 1}\right) = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$

If $\sinh \theta = 1$, then $\theta = \ln\left(1 + \sqrt{1^2 + 1}\right) = \ln\left(1 + \sqrt{2}\right)$

So $\theta = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$ or $\theta = \ln\left(1 + \sqrt{2}\right)$

40) d)

$2\cosh 2\theta \operatorname{sech} \theta = 7$

$\Rightarrow 2(2\cosh^2 \theta - 1) \times \frac{1}{\cosh \theta} = 7$

$\Rightarrow 2(2\cosh^2 \theta - 1) = 7\cosh \theta$

$\Rightarrow 4\cosh^2 \theta - 7\cosh \theta - 2 = 0$

$\Rightarrow \cosh \theta = -\frac{1}{4}$ or $\cosh \theta = 2$

$\cosh \theta = -\frac{1}{4}$ has no real solutions

If $\cosh \theta = 2$, then $\theta = \pm\ln\left(2 + \sqrt{2^2 - 1}\right) = \pm\ln\left(2 + \sqrt{3}\right)$

So $\theta = \ln\left(2 + \sqrt{3}\right)$ or $\theta = \ln\left(2 - \sqrt{3}\right)$

41) a)

$5\tanh 2x - 9\tanh x = 0$

$\Rightarrow 5 \times \frac{2\tanh x}{1+\tanh^2 x} - 9\tanh x = 0$

$\Rightarrow 10\tanh x - 9\tanh x\,(1+\tanh^2 x) = 0$

$\Rightarrow \tanh x\left(10 - 9(1+\tanh^2 x)\right) = 0$

$\Rightarrow \tanh x\,(1 - 9\tanh^2 x) = 0$

$\Rightarrow \tanh x = 0$ or $\tanh^2 x = \frac{1}{9} \Rightarrow \tanh x = \pm\frac{1}{3}$

If $\tanh x = 0$, then $x = \frac{1}{2}\ln\left(\frac{1+0}{1-0}\right) = \frac{1}{2}\ln 1 = 0$

If $\tanh x = \frac{1}{3}$, then $x = \frac{1}{2}\ln\left(\frac{1+\frac{1}{3}}{1-\frac{1}{3}}\right) = \frac{1}{2}\ln 2$

If $\tanh x = -\frac{1}{3}$, then $x = \frac{1}{2}\ln\left(\frac{1+\left(-\frac{1}{3}\right)}{1-\left(-\frac{1}{3}\right)}\right) = \frac{1}{2}\ln\frac{1}{2} = -\frac{1}{2}\ln 2$

So $x = 0$, $x = \frac{1}{2}\ln 2$ or $x = -\frac{1}{2}\ln 2$

41) b)

$4\tanh x - 3\,\text{tabnh}\,2x = 0$

$\Rightarrow 4\tanh x - 3 \times \frac{2\tanh x}{1+\tanh^2 x} = 0$

$\Rightarrow 4\tanh x\,(1+\tanh^2 x) - 6\tanh x = 0$

$\Rightarrow 2\tanh x\,(1+\tanh^2 x) - 3\tanh x = 0$

$\Rightarrow \tanh x\,(2(1+\tanh^2 x) - 3) = 0$

$\Rightarrow \tanh x\,(2\tanh^2 x - 1) = 0$

$\Rightarrow \tanh x = 0$ or $2\tanh^2 x - 1 = 0 \Rightarrow \tanh x = \pm\frac{1}{\sqrt{2}} = \pm\frac{\sqrt{2}}{2}$

If $\tanh x = 0$, then $x = \frac{1}{2}\ln\left(\frac{1+0}{1-0}\right) = \frac{1}{2}\ln 1 = 0$

If $\tanh x = \frac{\sqrt{2}}{2}$, then $x = \frac{1}{2}\ln\left(\frac{1+\frac{\sqrt{2}}{2}}{1-\frac{\sqrt{2}}{2}}\right) = \frac{1}{2}\ln\left(\frac{2+\sqrt{2}}{2-\sqrt{2}}\right) = \frac{1}{2}\ln\left(3+2\sqrt{2}\right)$

If $\tanh x = -\frac{\sqrt{2}}{2}$, then $x = \frac{1}{2}\ln\left(\frac{1+\left(-\frac{\sqrt{2}}{2}\right)}{1-\left(-\frac{\sqrt{2}}{2}\right)}\right) = \frac{1}{2}\ln\left(\frac{2-\sqrt{2}}{2+\sqrt{2}}\right) = \frac{1}{2}\ln\left(3-2\sqrt{2}\right)$

So $x = 0$, $x = \frac{1}{2}\ln\left(3+2\sqrt{2}\right)$ or $x = \frac{1}{2}\ln\left(3-2\sqrt{2}\right)$

42)

$\sinh x = \frac{4}{3}$

$\cosh^2 x - \sinh^2 x = 1 \Rightarrow \cosh^2 x - \left(\frac{4}{3}\right)^2 = 1$

$\Rightarrow \cosh^2 x = \frac{25}{9}$

$\Rightarrow \cosh x = \frac{5}{3}$ or $\cosh x = -\frac{5}{3}$

$\cosh x = -\frac{5}{3}$ has no real solutions, so $\cosh x = \frac{5}{3}$ only

$\therefore \sinh 2x = 2\sinh x \cosh x = 2 \times \frac{4}{3} \times \frac{5}{3} = \frac{40}{9}$

43)

$\sinh x = -\frac{1}{3}$

$\therefore \cosh 2x = 1 + 2\sinh^2 x = 1 + 2 \times \left(-\frac{1}{3}\right)^2 = \frac{11}{9}$

44)

$\cosh x = \frac{9}{4}$

$\cosh^2 x - \sinh^2 x = 1 \Rightarrow \left(\frac{9}{4}\right)^2 - \sinh^2 x = 1$

$\Rightarrow \sinh^2 x = \frac{65}{16}$

$\Rightarrow \sinh x = \frac{\sqrt{65}}{4}$ or $\sinh x = -\frac{\sqrt{65}}{4}$

$\tanh x = \frac{\sinh x}{\cosh x} = \frac{\frac{\sqrt{65}}{4}}{\frac{9}{4}} = \frac{\sqrt{65}}{9}$ or $\tanh x = \frac{\sinh x}{\cosh x} = \frac{-\frac{\sqrt{65}}{4}}{\frac{9}{4}} = -\frac{\sqrt{65}}{9}$

$\therefore \tanh 2x = \frac{2\tanh x}{1+\tanh^2 x} = \frac{2\times\frac{\sqrt{65}}{9}}{1+\left(\frac{\sqrt{65}}{9}\right)^2} = \frac{9\sqrt{65}}{73}$ or $\tanh 2x = \frac{2\tanh x}{1+\tanh^2 x} = \frac{2\times\left(-\frac{\sqrt{65}}{9}\right)}{1+\left(-\frac{\sqrt{65}}{9}\right)^2} = -\frac{9\sqrt{65}}{73}$

So $\tanh 2x = \pm\frac{9\sqrt{65}}{73}$

45) a)

$\sinh 3\theta \equiv \sinh(2\theta + \theta)$

$\equiv \sinh 2\theta \cosh\theta + \cosh 2\theta \sinh\theta$

$\equiv (2\sinh\theta\cosh\theta)\cosh\theta + (1 + 2\sinh^2\theta)\sinh\theta$

$\equiv 2\sinh\theta\cosh^2\theta + \sinh\theta + 2\sinh^3\theta$

$\equiv 2\sinh\theta(1 + \sinh^2\theta) + \sinh\theta + 2\sinh^3\theta$

$\equiv 2\sinh\theta + 2\sinh^3\theta + \sinh\theta + 2\sinh^3\theta$

$\equiv 4\sinh^3\theta + 3\sinh\theta$

45) b)

$20x^3 + 15x - 12 = 0$

$\Rightarrow 20\sinh^3\theta + 15\sinh\theta - 12 = 0$

$\Rightarrow 5(4\sinh^2\theta + 3\sinh\theta) - 12 = 0$

$\Rightarrow 5\sinh 3\theta - 12 = 0$

$\Rightarrow \sinh 3\theta = \frac{12}{5}$

$\Rightarrow 3\theta = \ln\left(\frac{12}{5} + \sqrt{\left(\frac{12}{5}\right)^2 + 1}\right)$

$\Rightarrow \theta = \frac{1}{3}\ln\left(\frac{12}{5} + \frac{13}{5}\right) = \frac{1}{3}\ln 5$

45) c)

$x = \sinh\theta$

$\Rightarrow x = \sinh\left(\frac{1}{3}\ln 5\right)$

$= \frac{e^{\frac{1}{3}\ln 5} - e^{-\frac{1}{3}\ln 5}}{2}$

$= \frac{e^{\ln 5^{\frac{1}{3}}} - e^{\ln 5^{-\frac{1}{3}}}}{2}$

$= \frac{1}{2}\left(5^{\frac{1}{3}} - 5^{-\frac{1}{3}}\right)$

46) a)

LHS: $\coth^2 x - \tanh^2 x$

$\equiv (\operatorname{cosech}^2 x + 1) - (1 - \operatorname{sech}^2 x)$

$\equiv \operatorname{cosech}^2 x + \operatorname{sech}^2 x$:RHS

46) b)

RHS: $\sinh^2 x - \tanh^2 x$

$\equiv \sinh^2 x - \frac{\sinh^2 x}{\cosh^2 x}$

$\equiv \sinh^2 x \left(1 - \frac{1}{\cosh^2 x}\right)$

$\equiv \sinh^2 x \, (1 - \operatorname{sech}^2 x)$

$\equiv \sinh^2 x \tanh^2 x$:LHS

46) c)

LHS: $\operatorname{cosech}^2 x - \operatorname{sech}^2 x$

$\equiv \frac{1}{\sinh^2 x} - \frac{1}{\cosh^2 x}$

$\equiv \frac{\cosh^2 x - \sinh^2 x}{\sinh^2 x \cosh^2 x}$

$\equiv \frac{1}{\sinh^2 x \cosh^2 x}$

$\equiv \frac{1}{(\sinh x \cosh x)^2}$

$\equiv \frac{4}{(2 \sinh x \cosh x)^2}$

$\equiv \frac{4}{(\sinh 2x)^2}$

$\equiv \frac{4}{\sinh^2 2x}$

$\equiv 4 \operatorname{cosech}^2 2x$:RHS

47) a)

LHS: $(\operatorname{sech} x - \tanh x)^2$

$\equiv \operatorname{sech}^2 x - 2 \operatorname{sech} x \tanh x + \tanh^2 x$

$\equiv (1 - \tanh^2 x) - 2 \operatorname{sech} x \tanh x + \tanh^2 x$

$\equiv 1 - 2 \operatorname{sech} x \tanh x$:RHS

47) b)

LHS: $(\operatorname{sech} x - \coth x)^2$

$\equiv \operatorname{sech}^2 x - 2 \operatorname{sech} x \coth x + \coth^2 x$

$\equiv (1 - \tanh^2 x) - 2 \times \frac{1}{\cosh x} \times \frac{\cosh x}{\sinh x} + (\operatorname{cosech}^2 x + 1)$

$\equiv 1 - \tanh^2 x - 2 \times \frac{1}{\sinh x} + \operatorname{cosech}^2 x + 1$

$\equiv 2 - \tanh^2 x - 2 \operatorname{cosec} x + \operatorname{cosech}^2 x$:RHS

47) c)

LHS: $(\operatorname{sech} x - \operatorname{cosech} x)^2$

$\equiv \left(\frac{1}{\cosh x} - \frac{1}{\sinh x}\right)^2$

$\equiv \left(\frac{\sinh x - \cosh x}{\sinh x \cosh x}\right)^2$

$\equiv \frac{(\sinh x - \cosh x)^2}{\sinh^2 x \cosh^2 x}$

$\equiv \operatorname{cosech}^2 x \operatorname{sech}^2 x (\sinh x - \cosh x)^2$

$\equiv \operatorname{cosech}^2 x \operatorname{sech}^2 x (\cosh^2 x + \sinh^2 x - 2\sinh x \cosh x)^2$

$\equiv \operatorname{cosech}^2 x \operatorname{sech}^2 x (\cosh 2x - \sinh 2x)^2$:RHS

48) a)

LHS: $\frac{1+\cosh x}{\sinh x} - \frac{\sinh x}{1+\cosh x}$

$\equiv \frac{(1+\cosh x)^2 - \sinh^2 x}{\sinh x(1+\cosh x)}$

$\equiv \frac{1+2\cosh x+\cosh^2 x-\sinh^2 x}{\sinh x(1+\cosh x)}$

$\equiv \frac{1+2\cosh x+1}{\sinh x(1+\cosh x)}$

$\equiv \frac{2(1+\cosh x)}{\sinh x(1+\cosh x)}$

$\equiv \frac{2}{\sinh x}$

$\equiv 2\operatorname{cosech} x$:RHS

48) b)

$\frac{1+\cosh t}{\sinh t} - \frac{\sinh t}{1+\cosh t} = -4$

$\Rightarrow 2\operatorname{cosech} t = -4$

$\Rightarrow \operatorname{cosech} t = -2$

$\Rightarrow \sinh t = -\frac{1}{2}$

$\Rightarrow t = \ln\left(-\frac{1}{2} + \sqrt{\left(-\frac{1}{2}\right)^2 + 1}\right) = \ln\left(-\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$

49) a)

LHS: $\frac{\sinh x}{\cosh x-1}-\operatorname{cosech} x$

$\equiv \frac{\sinh x}{\cosh x-1}\times\frac{\cosh x+1}{\cosh x+1}-\frac{1}{\sinh x}$

$\equiv \frac{\sinh x\cosh x+\sinh x}{\cosh^2 x-1}-\frac{1}{\sinh x}$

$\equiv \frac{\sinh x\cosh x+\sinh x}{\sinh^2 x}-\frac{\sinh x}{\sinh^2 x}$

$\equiv \frac{\sinh x\cosh x+\sinh x-\sinh x}{\sinh^2 x}$

$\equiv \frac{\sinh x\cosh x}{\sinh^2 x}$

$\equiv \frac{\cosh x}{\sinh x}$

$\equiv \coth x$:RHS

49) b)

$\frac{\sinh t}{\cosh t-1}=8+\coth t$

$\Rightarrow \coth t+\operatorname{cosech} t=8+\coth t$

$\Rightarrow \operatorname{cosech} t=8$

$\Rightarrow \sinh t=\frac{1}{8}$

$\Rightarrow t=\ln\left(\frac{1}{8}+\sqrt{\left(\frac{1}{8}\right)^2+1}\right)=\ln\left(\frac{1}{8}+\frac{\sqrt{65}}{8}\right)$

50)

$\operatorname{sech}\frac{x}{2}=6\sinh\frac{x}{2}$

$\Rightarrow \frac{1}{\cosh\frac{x}{2}}=6\sinh\frac{x}{2}$

$\Rightarrow 1=6\sinh\frac{x}{2}\cosh\frac{x}{2}$

$\Rightarrow 1=3\sinh x$

$\Rightarrow \sinh x=\frac{1}{3}$

$\Rightarrow x=\ln\left(\frac{1}{3}+\frac{\sqrt{10}}{3}\right)=\ln\left(\frac{1+\sqrt{10}}{3}\right)=\ln\left(1+\sqrt{10}\right)-\ln 3$ as required

51) a)

LHS: $\coth 2x - \tanh 2x$

$\equiv \frac{1}{\tanh 2x} - \tanh 2x$

$\equiv \frac{1-\tanh^2 2x}{\tanh 2x}$

$\equiv \frac{\operatorname{sech}^2 2x}{\tanh 2x}$

$\equiv \frac{\frac{1}{\cosh^2 2x}}{\frac{\sinh 2x}{\cosh 2x}}$

$\equiv \frac{1}{\cosh^2 2x} \times \frac{\cosh 2x}{\sinh 2x}$

$\equiv \frac{1}{\cosh 2x \sinh 2x}$

$\equiv \frac{2}{2\sinh 2x \cosh 2x}$

$\equiv \frac{2}{\sinh 4x}$

$\equiv 2\operatorname{cosech} 4x$:RHS

51) b)

LHS: $\coth x - \tanh 2x$

$\equiv \frac{1}{\tanh x} - \frac{2\tanh x}{1+\tanh^2 x}$

$\equiv \frac{1+\tanh^2 x-2\tanh^2 x}{\tanh x(1+\tanh^2 x)}$

$\equiv \frac{1-\tanh^2 x}{\tanh x(1+\tanh^2 x)}$

$\equiv \frac{\operatorname{sech}^2 x}{\tanh x(1+\tanh^2 x)}$

$\equiv \frac{\frac{1}{\cosh^2 x}}{\frac{\sinh x}{\cosh x}\left(1+\frac{\sinh^2 x}{\cosh^2 x}\right)}$

$\equiv \frac{\frac{1}{\cosh^2 x}}{\frac{\sinh x}{\cosh x}\left(1+\frac{\sinh^2 x}{\cosh^2 x}\right)} \times \frac{\cosh^2 x}{\cosh^2 x}$

$\equiv \frac{1}{\frac{\sinh x}{\cosh x}(\cosh^2 x+\sinh^2 x)}$

$\equiv \frac{\cosh x}{\sinh x} \times \frac{1}{\cosh^2 x+\sinh^2 x}$

$\equiv \coth x \times \frac{1}{\cosh 2x}$

$\equiv \coth x \operatorname{sech} 2x$:RHS

52) a)

LHS: $\coth 2x - \cosh 2x$

$\equiv \frac{\cosh 2x}{\sinh 2x} - \cosh 2x$

$\equiv \cosh 2x \left(\frac{1}{\sinh 2x} - 1\right)$

$\equiv \cosh 2x \left(\frac{1-\sinh 2x}{\sinh 2x}\right)$

$\equiv \frac{\cosh 2x}{\sinh 2x} \times (1 - \sinh 2x)$

$\equiv \coth 2x\,(1 - \sinh 2x)$: RHS

52) b)

LHS: $\coth 2x + \operatorname{cosech} 2x$

$\equiv \frac{\cosh 2x}{\sinh 2x} + \frac{1}{\sinh 2x}$

$\equiv \frac{\cosh 2x+1}{\sinh 2x}$

$\equiv \frac{2\cosh^2 x-1+1}{2\sinh x \cosh x}$

$\equiv \frac{2\cosh^2 x}{2\sinh x \cosh x}$

$\equiv \frac{\cosh x}{\sinh x}$

$\equiv \coth x$:RHS

53)

RHS: $\tanh 2x + \operatorname{cosech} 2x$

$\equiv \frac{\sinh 2x}{\cosh 2x} + \frac{1}{\sinh 2x}$

$\equiv \frac{\sinh^2 2x+\cosh 2x}{\sinh 2x \cosh 2x}$

$\equiv \frac{1}{\sinh 2x \cosh 2x} \times (\sinh^2 2x + \cosh 2x)$

$\equiv \frac{2}{2\sinh 2x \cosh 2x} \times (\sinh^2 2x + \cosh 2x)$

$\equiv \frac{2}{\sinh 4x} \times (\sinh^2 2x + \cosh 2x)$

$\equiv 2\operatorname{cosech} 4x\,(\sinh^2 2x + \cosh 2x)$:LHS

54) a)

LHS: $(\sinh 2x + \cosh 2x)^2$

$\equiv \sinh^2 2x + 2\sinh 2x \cosh 2x + \cosh^2 2x$

$\equiv 2\sinh 2x \cosh 2x + (\cosh^2 2x + \sinh^2 2x)$

$\equiv \sinh 4x + \cosh 4x$:RHS

54) b)

LHS: $\frac{2\sinh x}{\cosh 2x - 1} - \frac{2\sinh x}{\cosh 2x + 1}$

$\equiv \frac{2\sinh x(\cosh 2x+1) - 2\sinh x(\cosh 2x - 1)}{(\cosh 2x - 1)(\cosh 2x + 1)}$

$\equiv \frac{2\sinh x \cosh 2x + 2\sinh x - 2\sinh x \cosh 2x + 2\sinh x}{\cosh^2 2x - 1}$

$\equiv \frac{4\sinh x}{\cosh^2 2x - 1}$

$\equiv \frac{4\sinh x}{\sinh^2 2x}$

$\equiv \frac{4\sinh x}{(2\sinh x \cosh x)^2}$

$\equiv \frac{4\sinh x}{4\sinh^2 x \cosh^2 x}$

$\equiv \frac{1}{\sinh x \cosh^2 x}$

$\equiv \frac{1}{\sinh x} \times \frac{1}{\cosh^2 x}$

$\equiv \operatorname{cosech} x \operatorname{sech}^2 x$:RHS

55) a)

RHS: $\frac{1}{(\sinh x + \cosh x)^2}$

$\equiv \frac{1}{(\sinh x + \cosh x)^2} \times \frac{(\sinh x - \cosh x)^2}{(\sinh x - \cosh x)^2}$

$\equiv \frac{(\sinh x - \cosh x)^2}{(\sinh x + \cosh x)(\sinh x - \cosh x)(\sinh x + \cosh x)(\sinh x - \cosh x)}$

$\equiv \frac{(\sinh x - \cosh x)^2}{(\sinh^2 x - \cosh^2 x)(\sinh^2 x - \cosh^2 x)}$

$\equiv \frac{(\sinh x - \cosh x)^2}{1 \times 1}$

$\equiv (\sinh x - \cosh x)^2$

$\equiv \sinh^2 x - 2\sinh x \cosh x + \cosh^2 x$

$\equiv (\cosh^2 x + \sinh^2 x) - 2\sinh x \cosh x$

$\equiv \cosh 2x - \sinh 2x$:LHS

55) b)

RHS: $\frac{\cosh x(\coth x-1)}{\sinh x+\cosh x}$

$\equiv \frac{\cosh x\left(\frac{\cosh x}{\sinh x}-1\right)}{\sinh x+\cosh x}$

$\equiv \frac{\frac{\cosh x}{\sinh x}(\cosh x-\sinh x)}{\sinh x+\cosh x}$

$\equiv \frac{\cosh x}{\sinh x}\times\frac{\cosh x-\sinh x}{\sinh x+\cosh x}$

$\equiv \frac{\cosh x}{\sinh x}\times\frac{\cosh x-\sinh x}{\cosh x+\sinh x}\times\frac{\cosh x-\sinh x}{\cosh x-\sinh x}$

$\equiv \frac{\cosh x}{\sinh x}\times\frac{\cosh^2 x-2\sinh x\cosh x+\sinh^2 x}{\cosh^2 x-\sinh^2 x}$

$\equiv \frac{\cosh x}{\sinh x}\times\frac{(\cosh^2 x+\sinh^2 x)-2\sinh x\cosh x}{1}$

$\equiv \coth x\times\frac{\cosh 2x-\sinh 2x}{1}$

$\equiv \coth x\,(\cosh 2x-\sinh 2x)$:LHS

56)

$3\sinh 2x-8\sinh x-15\cosh x+20=0$

$\Rightarrow 6\sinh x\cosh x-8\sinh x-15\cosh x+20=0$

$\Rightarrow (2\sinh x-5)(3\cosh x-4)=0$

$\Rightarrow \sinh x=\frac{5}{2}$ or $\cosh x=\frac{4}{3}$

If $\sinh x=\frac{5}{2}$, then $x=\ln\left(\frac{5}{2}+\sqrt{\left(\frac{5}{2}\right)^2+1}\right)=\ln\left(\frac{5}{2}+\frac{\sqrt{29}}{2}\right)$

If $\cosh x=\frac{4}{3}$, then $x=\pm\ln\left(\frac{4}{3}+\sqrt{\left(\frac{4}{3}\right)^2-1}\right)=\pm\ln\left(\frac{4}{3}+\frac{\sqrt{7}}{3}\right)$

So $x=\ln\left(\frac{5}{2}+\frac{\sqrt{29}}{2}\right)$, $x=\ln\left(\frac{4}{3}+\frac{\sqrt{7}}{3}\right)$, or $\ln\left(\frac{4}{3}-\frac{\sqrt{7}}{3}\right)$

57) a)

$\sinh(\ln(8+2x)) = x$

$\Rightarrow \ln(8+2x) = \operatorname{arsinh} x$

$\Rightarrow \ln(8+2x) = \ln\left(x+\sqrt{x^2+1}\right)$

$\Rightarrow 8+2x = x+\sqrt{x^2+1}$

$\Rightarrow 8+x = \sqrt{x^2+1}$

$\Rightarrow (8+x)^2 = x^2+1$

$\Rightarrow x^2+16x+64 = x^2+1$

$\Rightarrow 16x = -63$

$\Rightarrow x = -\frac{63}{16}$

57) b)

$\cosh(\ln(3-5x)) + 10x = 0$

$\Rightarrow \cosh(\ln(3-5x)) = -10x$

$\Rightarrow \ln(3-5x) = \operatorname{arcosh}(-10x)$

$\Rightarrow \ln(3-5x) = \ln\left(-10x+\sqrt{(-10x)^2-1}\right)$

$\Rightarrow 3-5x = -10x+\sqrt{100x^2-1}$

$\Rightarrow 3+5x = \sqrt{100x^2-1}$

$\Rightarrow (3+5x)^2 = 100x^2-1$

$\Rightarrow 25x^2+30x+9 = 100x^2-1$

$\Rightarrow -75x^2+30x+10 = 0$

$\Rightarrow 15x^2-6x-2 = 0$

$\Rightarrow x = \frac{3\pm\sqrt{39}}{15}$

However, $x = \frac{3+\sqrt{39}}{15}$ cannot be a solution as $3-5\left(\frac{3+\sqrt{39}}{15}\right) = \frac{6-\sqrt{39}}{3} < 0$ and we cannot take the logarithm of a negative number.

So $x = \frac{3-\sqrt{39}}{15}$ is the only real solution.

57) c)

$\operatorname{sech}(\ln(4x+9)) + \frac{1}{12x} = 0$

$\Rightarrow \operatorname{sech}(\ln(4x+9)) = -\frac{1}{12x}$

$\Rightarrow \cosh(\ln(4x+9)) = -12x$

$\Rightarrow \ln(4x+9) = \operatorname{arcosh}(-12x)$

$\Rightarrow \ln(4x+9) = \ln\left(-12x + \sqrt{(-12x)^2 - 1}\right)$

$\Rightarrow 4x + 9 = -12x + \sqrt{144x^2 - 1}$

$\Rightarrow 16x + 9 = \sqrt{144x^2 - 1}$

$\Rightarrow (16x+9)^2 = 144x^2 - 1$

$\Rightarrow 256x^2 + 288x + 81 = 144x^2 - 1$

$\Rightarrow 112x^2 + 288x + 82 = 0$

$\Rightarrow x = \frac{-36 \pm 19\sqrt{2}}{28}$

57) d)

$\tanh\left(\ln\left(\sqrt{3+4x}\right)\right) = 2x$

$\Rightarrow \ln\left(\sqrt{3+4x}\right) = \operatorname{artanh} 2x$

$\Rightarrow \ln(3+4x)^{\frac{1}{2}} = \frac{1}{2}\ln\left(\frac{1+2x}{1-2x}\right)$

$\Rightarrow \frac{1}{2}\ln(3+4x) = \frac{1}{2}\ln\left(\frac{1+2x}{1-2x}\right)$

$\Rightarrow \ln(3+4x) = \ln\left(\frac{1+2x}{1-2x}\right)$

$\Rightarrow 3 + 4x = \frac{1+2x}{1-2x}$

$\Rightarrow (3+4x)(1-2x) = 1 + 2x$

$\Rightarrow -8x^2 - 2x + 3 = 1 + 2x$

$\Rightarrow -8x^2 - 4x + 2 = 0$

$\Rightarrow 4x^2 + 2x - 1 = 0$

$\Rightarrow x = \frac{-1 \pm \sqrt{5}}{4}$

However, $x = \frac{-1-\sqrt{5}}{4}$ cannot be a solution as $3 + 4\left(\frac{-1-\sqrt{5}}{4}\right) = 2 - \sqrt{5} < 0$ and we cannot take the square root of a negative number.

So $x = \frac{-1+\sqrt{5}}{4}$ is the only real solution.